CHAPTER 5

VECTOR SPACES

A large body of physical theory can be cast within the mathematical framework of vector spaces. Vector spaces are far more general than vectors in ordinary space, and the analogy may to the uninitiated seem somewhat strained. Basically, this subject deals with quantities that can be represented by expansions in a series of functions, and includes the methods by which such expansions can be generated and used for various purposes. A key aspect of the subject is the notion that a more or less arbitrary **function** can be represented by such an expansion, and that the coefficients in these expansions have transformation properties similar to those exhibited by vector components in ordinary space. Moreover, **operators** can be introduced to describe the application of various processes to a function, thereby converting it (and also the coefficients defining it) into other functions within our vector space. The concepts presented in this chapter are crucial to an understanding of quantum mechanics, to classical systems involving oscillatory motion, transport of material or energy, even to fundamental particle theory. Indeed, it is not excessive to claim that vector spaces are one of the most fundamental mathematical structures in physical theory.

5.1 VECTORS IN FUNCTION SPACES

We now seek to extend the concepts of classical vector analysis (from Chapter 3) to more general situations. Suppose that we have a two-dimensional (2-D) space in which the two coordinates, which are real (or in the most general case, complex) numbers that we will call a_1 and a_2, are, respectively, associated with the two functions $\varphi_1(s)$ and $\varphi_2(s)$. It is important at the outset to understand that our new 2-D space has nothing whatsoever to do with the physical xy space. It is a space in which the coordinate point (a_1, a_2) corresponds to the function

$$f(s) = a_1\varphi_1(s) + a_2\varphi_2(s). \tag{5.1}$$

The analogy with a physical 2-D vector space with vectors $\mathbf{A} = A_1\hat{\mathbf{e}}_1 + A_2\hat{\mathbf{e}}_2$ is that $\varphi_i(s)$ corresponds to $\hat{\mathbf{e}}_i$, while $a_i \longleftrightarrow A_i$, and $f(s) \longleftrightarrow \mathbf{A}$. In other words, the coordinate

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values are the **coefficients** of the $\varphi_i(s)$, so each point in the space identifies a different function $f(s)$. Both f and φ are shown above as dependent on an independent variable we call s. We choose the name s to emphasize the fact that the formulation is not restricted to the spatial variables x, y, z, but can be whatever variable, or set of variables, is needed for the problem at hand. Note further that the variable s is not a continuous analog of the discrete variables x_i of an ordinary vector space. It is a parameter reminding the reader that the φ_i that correspond to the dimensions of our vector space are usually not just numbers, but are functions of one or more variables. The variable(s) denoted by s may sometimes correspond to physical displacements, but that is not always the case. What should be clear is that s has nothing to do with the coordinates in our vector space; that is the role of the a_i.

Equation (5.1) defines a set of functions (a **function space**) that can be built from the **basis** φ_1, φ_2; we call this space a **linear vector space** because its members are linear combinations of the basis functions and the addition of its members corresponds to component (coefficient) addition. If $f(s)$ is given by Eq. (5.1) and $g(s)$ is given by another linear combination of **the same** basis functions,

$$g(s) = b_1\varphi_1(s) + b_2\varphi_2(s),$$

with b_1 and b_2 the coefficients defining $g(s)$, then

$$h(s) = f(s) + g(s) = (a_1 + b_1)\varphi_1(s) + (a_2 + b_2)\varphi_2(s) \tag{5.2}$$

defines $h(s)$, the member of our space (i.e., the function), which is the sum of the members $f(s)$ and $g(s)$. In order for our vector space to be useful, we consider only spaces in which the sum of any two members of the space is also a member.

In addition, the notion of linearity includes the requirement that if $f(s)$ is a member of our vector space, then $u(s) = k\,f(s)$, where k is a real or complex number, is also a member, and we can write

$$u(s) = k\,f(s) = ka_1\varphi_1(s) + ka_2\varphi_2(s). \tag{5.3}$$

Vector spaces for which addition of two members or multiplication of a member by scalar always produces a result that is also a member are termed **closed** under these operations.

We can summarize our findings up to this point as follows: addition of two members of our vector space causes the coefficients of the sum, $h(s)$ in Eq. (5.2), to be the sum of the coefficients of the addends, namely $f(s)$ and $g(s)$; multiplication of $f(s)$ by a ordinary number k (which, by analogy with ordinary vectors, we call a **scalar**), results in the multiplication of the coefficients by k. These are exactly the operations we would carry out to form the sum of two ordinary vectors, $\mathbf{A} + \mathbf{B}$, or the multiplication of a vector by a scalar, as in $k\mathbf{A}$. However, here we have the coefficients a_i and b_i, which combine under vector addition and multiplication by a scalar in exactly the same way that we would combine the ordinary vector components A_i and B_i.

The functions that form the basis of our vector space can be ordinary functions, and may be as simple as powers of s, or more complicated, as for example $\varphi_1 = (1 + 3s + 3s^2)e^s$, $\varphi_2 = (1 - 3s + 3s^2)e^{-s}$, or compound quantities such as the Pauli matrices σ_i, or even completely abstract quantities that are defined only by certain properties they may possess. The number of basis functions (i.e., the **dimension** of our basis) may be a small number such as 2 or 3, a larger but finite integer, or even denumerably infinite (as would arise in an untruncated power series). The main universal restriction on the form of a basis is that the basis members be linearly independent, so that any function (member) of our vector space will be described by a unique linear combination of the basis functions. We illustrate the possibilities with some simple examples.

Example 5.1.1 SOME VECTOR SPACES

1. We consider first a vector space of dimension 3, which is **spanned by** (meaning that it has a basis that consists of) the three functions $P_0(s) = 1$, $P_1(s) = s$, $P_2(s) = \frac{3}{2}s^2 - \frac{1}{2}$. Some members of this vector space include the functions

$$s + 3 = 3P_0(s) + P_1(s), \quad s^2 = \frac{1}{3}P_0(s) + \frac{2}{3}P_2(s), \quad 4 - 3s = 4P_0(s) - 3P_1(s).$$

In fact, because we can write 1, s, and s^2 in terms of our basis, we can see that **any** quadratic form in s will be a member of our vector space, and that our space includes only functions of s that can be written in the form $c_0 + c_1 s + c_2 s^2$.

To illustrate our vector-space operations, we can form

$$\begin{aligned} s^2 - 2(s+3) &= \left[\frac{1}{3}P_0(s) + \frac{2}{3}P_2(s)\right] - 2\Big[3P_0(s) + P_1(s)\Big] \\ &= \left(\frac{1}{3} - 6\right)P_0(s) - 2P_1(s) + \frac{2}{3}P_2(s). \end{aligned}$$

This calculation involves only operations on the coefficients; we do not need to refer to the definitions of the P_n to carry it out.

Note that we are free to define our basis any way we want, so long as its members are linearly independent. We could have chosen as our basis for this same vector space $\varphi_0 = 1$, $\varphi_1 = s$, $\varphi_2 = s^2$, but we chose not to do so.

2. The set of functions $\varphi_n(s) = s^n$ $(n = 0, 1, 2, \ldots)$ is a basis for a vector space whose members consist of functions that can be represented by a Maclaurin series. To avoid difficulties with this infinite-dimensional basis, we will usually need to restrict consideration to functions and ranges of s for which the Maclaurin series converges. Convergence and related issues are of great interest in pure mathematics; in physics problems we usually proceed in ways such that convergence is assured.

The members of our vector space will have representations

$$f(s) = a_0 + a_1 s + a_2 s^2 + \cdots = \sum_{n=0}^{\infty} a_n s^n,$$

and we can (at least in principle) use the rules for making power series expansions to find the coefficients that correspond to a given $f(s)$.

3. The spin space of an electron is spanned by a basis that consists of a linearly independent set of possible spin states. It is well known that an electron can have two linearly independent spin states, and they are often denoted by the symbols $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. One possible spin state is $f = a_1\boldsymbol{\alpha} + a_2\boldsymbol{\beta}$, and another is $g = b_1\boldsymbol{\alpha} + b_2\boldsymbol{\beta}$. We do not even need

to know what $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ really stand for to discuss the 2-D vector space spanned by these functions, nor do we need to know the role of any parametric variable such as s. We can, however, state that the particular spin state corresponding to $f+ig$ must have the form

$$f+ig=(a_1+ib_1)\boldsymbol{\alpha}+(a_2+ib_2)\boldsymbol{\beta}.$$

■

Scalar Product

To make the vector space concept useful and parallel to that of vector algebra in ordinary space, we need to introduce the concept of a scalar product in our function space. We shall write the scalar product of two members of our vector space, f and g, as $\langle f|g\rangle$. This is the notation that is almost universally used in physics; various other notations can be found in the mathematics literature; examples include $[f,g]$ and (f,g).

The scalar product has two main features, the full meaning of which may only become clear as we proceed. They are:

1. The scalar product of a member with itself, e.g., $\langle f|f\rangle$, must evaluate to a numerical value (not a function) that plays the role of the square of the magnitude of that member, corresponding to the dot product of an ordinary vector with itself, and
2. The scalar product must be linear in each of the two members.[1]

There exists an extremely wide range of possibilities for defining scalar products that meet these criteria. The situation that arises most often in physics is that the members of our vector space are ordinary functions of the variable s (as in the first vector space of Example 5.1.1), and the scalar product of the two members $f(s)$ and $g(s)$ is computed as an integral of the type

$$\langle f|g\rangle=\int_a^b f^*(s)g(s)\,w(s)\,ds, \tag{5.4}$$

with the choice of a, b, and $w(s)$ dependent on the particular definition we wish to adopt for our scalar product. In the special case $\langle f|f\rangle$, the scalar product is to be interpreted as the square of a "length," and this scalar product must therefore be positive for any f that is not itself identically zero. Since the integrand in the scalar product is then $f^*(s)f(s)w(s)$ and $f^*(s)f(s)\geq 0$ for all s (even if $f(s)$ is complex), we can see that $w(s)$ must be positive over the entire range $[a,b]$ except possibly for zeros at isolated points.

Let's review some of the implications of Eq. (5.4). It is not appropriate to interpret that equation as a continuum analog of the ordinary dot product, with the variable s thought of as the continuum limit of an index labeling vector components. The integral actually arises pursuant to a decision to compute a "squared length" as a possibly weighted average over the range of values of the parameter s. We can illustrate this point by considering the

[1] If the members of the vector space are complex, this statement will need adjustment; see the formal definitions in the next subsection.

other situation that arises occasionally in physics, and illustrated by the third vector space in Example 5.1.1. Here we simply **define** the scalar products of the individual $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ to have values

$$\langle\boldsymbol{\alpha}|\boldsymbol{\alpha}\rangle = \langle\boldsymbol{\beta}|\boldsymbol{\beta}\rangle = 1, \quad \langle\boldsymbol{\alpha}|\boldsymbol{\beta}\rangle = \langle\boldsymbol{\beta}|\boldsymbol{\alpha}\rangle = 0,$$

and then, taking the simple one-electron functions

$$f = a_1\boldsymbol{\alpha} + a_2\boldsymbol{\beta}, \quad g = b_1\boldsymbol{\alpha} + b_2\boldsymbol{\beta},$$

and assuming a_i and b_i to be real, we expand $\langle f|g\rangle$ (using its linearity property) to reach

$$\langle f|g\rangle = a_1 b_1\langle\boldsymbol{\alpha}|\boldsymbol{\alpha}\rangle + a_1 b_2\langle\boldsymbol{\alpha}|\boldsymbol{\beta}\rangle + a_2 b_1\langle\boldsymbol{\beta}|\boldsymbol{\alpha}\rangle + a_2 b_2\langle\boldsymbol{\beta}|\boldsymbol{\beta}\rangle = a_1 b_1 + a_2 b_2. \tag{5.5}$$

These equations show that the introduction of an integral is not an indispensible step toward generalization of the scalar product; they also show that the final formula in Eq. (5.5), which **is** analogous to ordinary vector algebra, arises from the expansion of $\langle f|g\rangle$ in a basis whose two members, $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, are orthogonal (i.e., have a zero scalar product). Thus, the analogy to ordinary vector algebra is that the "unit vectors" of this spin system define an orthogonal "coordinate system" and that the "dot product" then has the expected form.

Vector spaces that are closed under addition and multiplication by a scalar and which have a scalar product that exists for all pairs of its members are termed **Hilbert spaces**; these are the vector spaces of primary importance in physics.

Hilbert Space

Proceeding now somewhat more formally (but still without complete rigor), and including the possibility that our function space may require more than two basis functions, we identify a Hilbert space $\mathcal{H}$ as having the following properties:

- Elements (members) f, g, or h of $\mathcal{H}$ are subject to two operations, **addition**, and **multiplication by a scalar** (here k, k_1, or k_2). These operations produce quantities that are also members of the space.
- Addition is commutative and associative:

$$f(s) + g(s) = g(s) + f(s), \quad [f(s) + g(s)] + h(s) = f(s) + [g(s) + h(s)].$$

- Multiplication by a scalar is commutative, associative, and distributive:

$$k\,f(s) = f(s)\,k, \quad k[f(s) + g(s)] = kf(s) + kg(s),$$
$$(k_1 + k_2)f(s) = k_1 f(s) + k_2 f(s), \quad k_1[k_2 f(s)] = k_1 k_2 f(s).$$

- $\mathcal{H}$ is **spanned** by a set of basis functions φ_i, where for the purposes of this book the number of such basis functions (the range of i) can either be finite or denumerably infinite (like the positive integers). This means that every function in $\mathcal{H}$ can be represented by the linear form $f(s) = \sum_n a_n\varphi_n(s)$. This property is also known as **completeness**. We require that the basis functions be linearly independent, so that each function in the space will be a unique linear combination of the basis functions.

- For all functions $f(s)$ and $g(s)$ in $\mathcal{H}$, there exists a scalar product, denoted as $\langle f|g\rangle$, which evaluates to a finite real or complex numerical value (i.e., does not contain s) and which has the properties that

  1. $\langle f|f\rangle \geq 0$, with the equality holding only if f is identically zero.[2] The quantity $\langle f|f\rangle^{1/2}$ is called the **norm** of f and is written $||f||$.
  2. $\langle g|f\rangle^* = \langle f|g\rangle$, $\langle f|g+h\rangle = \langle f|g\rangle + \langle f|h\rangle$, and $\langle f|kg\rangle = k\langle f|g\rangle$.

  Consequences of these properties are that $\langle f|k_1 g + k_2 h\rangle = k_1\langle f|g\rangle + k_2\langle f|h\rangle$, but $\langle kf|g\rangle = k^*\langle f|g\rangle$ and $\langle k_1 f + k_2 g|h\rangle = k_1^*\langle f|h\rangle + k_2^*\langle g|h\rangle$.

Example 5.1.2 Some Scalar Products

Continuing with the first vector space of Example 5.1.1, let's assume that our scalar product of any two functions $f(s)$ and $g(s)$ takes the form

$$\langle f|g\rangle = \int_{-1}^{1} f^*(s)\, g(s)\, ds, \tag{5.6}$$

i.e., the formula given as Eq. (5.4) with $a = -1$, $b = 1$, and $w(s) = 1$. Since all the members of this vector space are quadratic forms and the integral in Eq. (5.6) is over the finite range from -1 to $+1$, the scalar product will always exist and our three basis functions indeed define a Hilbert space. Before we make a few sample computations, let's note that the brackets in the left member of Eq. (5.6) do not show the detailed form of the scalar product, thereby concealing information about the integration limits, the number of variables (here we have only one, s), the nature of the space involved, the presence or absence of a weight factor $w(s)$, and even the exact operation that forms the product. All these features must be inferred from the context or by a previously provided definition.

Now let's evaluate two scalar products:

$$\langle P_0|s^2\rangle = \int_{-1}^{1} P_0^*(s)s^2\, ds = \int_{-1}^{1} (1)(s^2)dx = \left[\frac{s^3}{3}\right]_{-1}^{1} = \frac{2}{3},$$

$$\langle P_0|P_2\rangle = \int_{-1}^{1} (1)\left[\frac{3}{2}s^2 - \frac{1}{2}\right]ds = \left[\frac{3}{2}\frac{s^3}{3} - \frac{1}{2}s\right]_{-1}^{1} = 0. \tag{5.7}$$

Looking further at the scalar product definition of the present example, we note that it is consistent with the general requirements for a scalar product, as (1) $\langle f|f\rangle$ is formed as the integral of an inherently nonnegative integrand, and will be positive for all nonzero

[2] To be rigorous, the phrase "identically zero" needs to be replaced by "zero except on a set of measure zero," and other conditions need to be more tightly specified. These are niceties that are important for a precise formulation of the mathematics but are not often of practical importance to the working physicist. We note, however, that discontinuous functions do arise in applications of Fourier series, with consequences that are discussed in Chapter 19.

f; and (2) the placement of the complex-conjugate asterisk makes it obvious that $\langle g|f\rangle^* = \langle f|g\rangle$. ■

Schwarz Inequality

Any scalar product that meets the Hilbert space conditions will satisfy the **Schwarz inequality**, which can be stated as

$$|\langle f|g\rangle|^2 \le \langle f|f\rangle\langle g|g\rangle. \tag{5.8}$$

Here there is equality only if f and g are proportional. In ordinary vector space, the equivalent result is, referring to Eq. (1.113),

$$(\mathbf{A}\cdot\mathbf{B})^2 = |\mathbf{A}|^2|\mathbf{B}|^2\ \cos^2\theta \le |\mathbf{A}|^2|\mathbf{B}|^2, \tag{5.9}$$

where θ is the angle between the directions of **A** and **B**. As observed previously, the equality only holds if **A** and **B** are collinear. If we also require **A** to be of unit length, we have the intuitively obvious result that the projection of **B** onto a noncollinear **A** direction will have a magnitude less than that of **B**. The Schwarz inequality extends this property to functions; their norms shrink on nontrivial projection.

The Schwarz inequality can be proved by considering

$$I = \langle f - \lambda g|f - \lambda g\rangle \ge 0, \tag{5.10}$$

where λ is an as yet undetermined constant. Treating λ and λ^* as linearly independent,[3] we differentiate I with respect to λ^* (remember that the left member of the product is complex conjugated) and set the result to zero, to find the λ value for which I is a minimum:

$$-\langle g|f - \lambda g\rangle = 0 \quad\Longrightarrow\quad \lambda = \frac{\langle g|f\rangle}{\langle g|g\rangle}.$$

Substituting this λ value into Eq. (5.10), we get (using properties of the scalar product)

$$\langle f|f\rangle - \frac{\langle f|g\rangle\langle g|f\rangle}{\langle g|g\rangle} \ge 0.$$

Noting that $\langle g|g\rangle$ must be positive, and rewriting $\langle g|f\rangle$ as $\langle f|g\rangle^*$, we confirm the Schwarz inequality, Eq. (5.8).

Orthogonal Expansions

With now a well-behaved scalar product in hand, we can make the definition that two functions f and g are **orthogonal** if $\langle f|g\rangle = 0$, which means that $\langle g|f\rangle$ will also vanish. An example of two functions that are orthogonal under the then-applicable definition of the scalar product are $P_0(s)$ and $P_2(s)$, where the scalar product is that defined in Eq. (5.6) and P_0, P_2 are the functions from Example 5.1.1; the orthogonality is shown by Eq. (5.7). We further define a function f as **normalized** if the scalar product $\langle f|f\rangle = 1$; this is the

[3] It is not obvious that one can do this, but consider $\lambda = \mu + i\nu$, $\lambda^* = \mu - i\nu$, with μ and ν real. Then $\frac{1}{2}[\partial/\partial\mu + i\partial/\partial\nu]$ is equivalent to taking $\partial/\partial\lambda^*$ keeping λ constant.

function-space equivalent of a unit vector. We will find that great convenience results if the basis functions for our function space are normalized and mutually orthogonal, corresponding to the description of a 2-D or three-dimensional (3-D) physical vector space based on orthogonal unit vectors. A set of functions that is both normalized and mutually orthogonal is called an **orthonormal** set. If a member f of an orthogonal set is not normalized, it can be made so without disturbing the orthogonality: we simply rescale it to $\overline{f} = f/\langle f|f\rangle^{1/2}$, so any orthogonal set can easily be made orthonormal if desired.

If our basis is orthonormal, the coefficients for the expansion of an arbitrary function in that basis take a simple form. We return to our 2-D example, with the assumption that the φ_i are orthonormal, and consider the result of taking the scalar product of $f(s)$, as given by Eq. (5.1), with $\varphi_1(s)$:

$$\langle\varphi_1|f\rangle = \langle\varphi_1|(a_1\varphi_1 + a_2\varphi_2)\rangle = a_1\langle\varphi_1|\varphi_1\rangle + a_2\langle\varphi_1|\varphi_2\rangle. \tag{5.11}$$

The orthonormality of the φ now comes into play; the scalar product multiplying a_1 is unity, while that multiplying a_2 is zero, so we have the simple and useful result $\langle\varphi_1|f\rangle = a_1$. Thus, we have a rather mechanical means of identifying the components of f. The general result corresponding to Eq. (5.11) follows:

$$\text{If } \langle\varphi_i|\varphi_j\rangle = \delta_{ij} \quad \text{and} \quad f = \sum_{i=1}^{n} a_i\varphi_i, \quad \text{then} \quad a_i = \langle\varphi_i|f\rangle\,. \tag{5.12}$$

Here the **Kronecker delta**, δ_{ij}, is unity if $i = j$ and zero otherwise. Looking once again at Eq. (5.11), we consider what happens if the φ_i are orthogonal but not normalized. Then instead of Eq. (5.12) we would have:

$$\text{If the } \varphi_i \text{ are orthogonal and } f = \sum_{i=1}^{n} a_i\varphi_i, \quad \text{then} \quad a_i = \frac{\langle\varphi_i|f\rangle}{\langle\varphi_i|\varphi_i\rangle}\,. \tag{5.13}$$

This form of the expansion will be convenient when normalization of the basis introduces unpleasant factors.

Example 5.1.3 Expansion in Orthonormal Functions

Consider the set of functions $\chi_n(x) = \sin nx$, for $n = 1, 2, \ldots$, to be used for x in the interval $0 \le x \le \pi$ with scalar product

$$\langle f|g\rangle = \int_0^\pi f^*(x)g(x)dx. \tag{5.14}$$

We wish to use these functions for the expansion of the function $x^2(\pi - x)$.

First, we check that they are orthogonal:

$$S_{nm} = \int_0^\pi \chi_n^*(x)\chi_m(x)dx = \int_0^\pi \sin nx \sin mx\, dx.$$

For $n \neq m$ this integral can be shown to vanish, either by symmetry considerations or by consulting a table of integrals. To determine normalization, we need S_{nn}; from symmetry considerations, the integrand, $\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$, can be seen to have average value 1/2 over the range $(0, \pi)$, leading to $S_{nn} = \pi/2$ for all integer n. This means the χ_n are not normalized, but can be made so if we multiply by $\sqrt{2/\pi}$. So our orthonormal basis will be

$$\varphi_n(x) = \left(\frac{2}{\pi}\right)^{1/2} \sin nx, \quad n = 1, 2, 3, \ldots. \tag{5.15}$$

To expand $x^2(\pi - x)$, we apply Eq. (5.2), which requires the evaluation of

$$a_n = \langle \varphi_n | x^2(\pi - x) \rangle = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\pi} (\sin nx)\, x^2(\pi - x)\, dx, \tag{5.16}$$

for use in the expansion

$$x^2(\pi - x) = \left(\frac{2}{\pi}\right)^{1/2} \sum_{n=0}^{\infty} a_n \sin nx. \tag{5.17}$$

Evaluating cases of Eq. (5.16) by hand or using a computer for symbolic computation, we have for the first few a_n: $a_1 = 5.0132$, $a_2 = -1.8300$, $a_3 = 0.1857$, $a_4 = -0.2350$. The convergence is not very fast. ■

Example 5.1.4 Spin Space

A system of four spin-$\frac{1}{2}$ particles in a triplet state has the following three linearly independent spin functions:

$$\chi_1 = \alpha\beta\alpha\alpha - \beta\alpha\alpha\alpha, \quad \chi_2 = \alpha\alpha\alpha\beta - \alpha\alpha\beta\alpha, \quad \chi_3 = \alpha\alpha\alpha\beta + \alpha\alpha\beta\alpha - \alpha\beta\alpha\alpha - \beta\alpha\alpha\alpha.$$

The four symbols in each term of these expressions refer to the spin assignments of the four particles, in numerical order.

The scalar product in the spin space has the form, for monomials,

$$\langle abcd | wxyz \rangle = \delta_{aw}\delta_{bx}\delta_{cy}\delta_{dz},$$

meaning that the scalar product is unity if the two monomials are identical, and is zero if they are not. Scalar products involving polynomials can be evaluated by expanding them into sums of monomial products. It is easy to confirm that this definition meets the requirements for a valid scalar product.

Our mission will be (1) verify that the χ_i are orthogonal; (2) convert them, if necessary, to normalized form to make an orthonormal basis for the spin space; and (3) expand the following triplet spin function as a linear combination of the orthonormal spin basis functions:

$$\chi_0 = \alpha\alpha\beta\alpha - \alpha\beta\alpha\alpha.$$

The functions χ_1 and χ_2 are orthogonal, as they have no terms in common. Although χ_1 and χ_3 have two terms in common, they occur in sign combinations leading to a vanishing scalar product. The same observation applies to $\langle\chi_2|\chi_3\rangle$. However, none of the χ_i are normalized. We find $\langle\chi_1|\chi_1\rangle = \langle\chi_2|\chi_2\rangle = 2$, $\langle\chi_3|\chi_3\rangle = 4$, so an orthonormal basis would be

$$\varphi_1 = 2^{-1/2}\chi_1, \quad \varphi_2 = 2^{-1/2}\chi_2, \quad \varphi_3 = \frac{1}{2}\chi_3.$$

Finally, we obtain the coefficients for the expansion of χ_0 by forming $a_1 = \langle\varphi_1|\chi_0\rangle = -1/\sqrt{2}$, $a_2 = \langle\varphi_2|\chi_0\rangle = -1/\sqrt{2}$, and $a_3 = \langle\varphi_3|\chi_0\rangle = 1$. Thus, the desired expansion is

$$\chi_0 = -\frac{1}{\sqrt{2}}\varphi_1 - \frac{1}{\sqrt{2}}\varphi_2 + \varphi_3.$$

■

Expansions and Scalar Products

If we have found the expansions of two functions,

$$f = \sum_\mu a_\mu \varphi_\mu \quad \text{and} \quad g = \sum_\nu b_\nu \varphi_\nu,$$

then their scalar product can be written

$$\langle f|g\rangle = \sum_{\mu\nu} a_\mu^* b_\nu \langle\varphi_\mu|\varphi_\nu\rangle.$$

If the φ set is orthonormal, the above reduces to

$$\langle f|g\rangle = \sum_\mu a_\mu^* b_\mu. \tag{5.18}$$

In the special case $g = f$, this reduces to

$$\langle f|f\rangle = \sum_\mu |a_\mu|^2, \tag{5.19}$$

consistent with the requirement that $\langle f|f\rangle \geq 0$, with equality only if f is zero "almost everywhere."

If we regard the set of expansion coefficients a_μ as the elements of a column vector $\mathbf{a}$ representing f, with column vector $\mathbf{b}$ similarly representing g, Eqs. (5.18) and (5.19) correspond to the matrix equations

$$\langle f|g\rangle = \mathbf{a}^\dagger\mathbf{b}, \quad \langle f|f\rangle = \mathbf{a}^\dagger\mathbf{a}. \tag{5.20}$$

Note that by taking the adjoint of $\mathbf{a}$, we both complex conjugate it and convert it into a row vector, so that the matrix products in Eq. (5.20) collapse to scalars, as required.

Example 5.1.5 Coefficient Vectors

A set of functions that is orthonormal on $0 \le x \le \pi$ is

$$\varphi_n(x) = \sqrt{\frac{2-\delta_{n0}}{\pi}}\,\cos nx, \quad n = 0, 1, 2, \ldots .$$

First, let us expand in terms of this basis the two functions

$$\psi_1 = \cos^3 x + \sin^2 x + \cos x + 1 \quad \text{and} \quad \psi_2 = \cos^2 x - \cos x .$$

We write the expansions as vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ with components $n = 0, \ldots, 3$:

$$\mathbf{a}_1 = \begin{pmatrix} \langle\varphi_0|\psi_1\rangle \\ \langle\varphi_1|\psi_1\rangle \\ \langle\varphi_2|\psi_1\rangle \\ \langle\varphi_3|\psi_1\rangle \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} \langle\varphi_0|\psi_2\rangle \\ \langle\varphi_1|\psi_2\rangle \\ \langle\varphi_2|\psi_2\rangle \\ \langle\varphi_3|\psi_2\rangle \end{pmatrix}.$$

All components beyond $n = 3$ vanish and need not be shown. It is straightforward to evaluate these scalar products. Alternatively, we can rewrite the ψ_i using trigonometric identities, reaching the forms

$$\psi_1 = \frac{\cos 3x}{4} - \frac{\cos 2x}{2} + \frac{7}{4}\cos x + \frac{3}{2}, \quad \psi_2 = \frac{\cos 2x}{2} - \cos x + \frac{1}{2}.$$

These expressions are now easily recognized as equivalent to

$$\psi_1 = \sqrt{\frac{\pi}{2}}\left(\frac{\varphi_3}{4} - \frac{\varphi_2}{2} + \frac{7\varphi_1}{4} + \frac{3\sqrt{2}\,\varphi_0}{2}\right), \quad \psi_2 = \sqrt{\frac{\pi}{2}}\left(\frac{\varphi_2}{2} - \varphi_1 + \frac{\sqrt{2}\,\varphi_0}{2}\right),$$

so

$$\mathbf{a}_1 = \sqrt{\frac{\pi}{2}}\begin{pmatrix} 3\sqrt{2}/2 \\ 7/4 \\ -1/2 \\ 1/4 \end{pmatrix}, \quad \mathbf{a}_2 = \sqrt{\frac{\pi}{2}}\begin{pmatrix} \sqrt{2}/2 \\ -1 \\ 1/2 \\ 0 \end{pmatrix}.$$

We see from the above that the general formula for finding the coefficients in an orthonormal expansion, Eq. (5.12), is a systematic way of doing what sometimes can be carried out in other ways.

We can now evaluate the scalar products $\langle\psi_i|\psi_j\rangle$. Identifying these first as matrix products that we then evaluate,

$$\langle\psi_1|\psi_1\rangle = \mathbf{a}_1^\dagger\mathbf{a}_1 = \frac{63\pi}{16}, \quad \langle\psi_1|\psi_2\rangle = \mathbf{a}_1^\dagger\mathbf{a}_2 = -\frac{\pi}{4}, \quad \langle\psi_2|\psi_2\rangle = \mathbf{a}_2^\dagger\mathbf{a}_2 = \frac{7\pi}{8}.$$

■

Bessel's Inequality

Given a set of basis functions and the definition of a space, it is not necessarily assured that the basis functions span the space (a property sometimes referred to as **completeness**). For example, we might have a space defined to be that containing all functions possessing a scalar product of a given definition, while the basis functions have been specified by giving their functional form. This issue is of some importance, because we need to know whether an attempt to expand a function in a given basis can be guaranteed to converge to the correct result. Totally general criteria are not available, but useful results have been obtained if the function being expanded has, at worst, a finite number of finite discontinuities, and results are accepted as "accurate" if deviations from the correct value occur only at isolated points. Power series and trigonometric series have been proved complete for the expansion of square integrable functions f (those for which $\langle f|f\rangle$ as defined in Eq. (5.7) exists; mathematicians identify such spaces by the designation $\mathcal{L}^2$). Also proved complete are the orthonormal sets of functions that arise as the solutions to Hermitian eigenvalue problems.[4]

A not too practical test for completeness is provided by **Bessel's inequality**, which states that if a function f has been expanded in an orthonormal basis as $\sum_n a_n \varphi_n$, then

$$\langle f|f\rangle \geq \sum_n |a_n|^2, \tag{5.21}$$

with the inequality occurring if the expansion of f is incomplete. The impracticality of this as a completeness test is that one needs to apply it for all f before using it to claim completeness of the space.

We establish Bessel's inequality by considering

$$I = \left\langle f - \sum_i a_i\,\varphi_i \,\middle|\, f - \sum_j a_j\,\varphi_j \right\rangle \geq 0, \tag{5.22}$$

where $I = 0$ represents what is termed **convergence in the mean**, a criterion that permits the integrand to deviate from zero at isolated points. Expanding the scalar product, and eliminating terms that vanish because the φ are orthonormal, we arrive at Eq. (5.21), with equality only resulting if the expansion converges to f. We note in passing that convergence in the mean is a less stringent requirement than **uniform convergence**, but is adequate for almost all physical applications of basis-set expansions.

Example 5.1.6 Expansion of a Discontinuous Function

The functions $\cos nx$ $(n = 0, 1, 2, \dots)$ and $\sin nx$ $(n = 1, 2, \dots)$ have (together) been shown to form a complete set on the interval $-\pi < x < \pi$. Since this determination is obtained subject to convergence in the mean, there is the possibility of deviation at isolated points, thereby permitting the description of functions with isolated discontinuities.

[4] See R. Courant and D. Hilbert, *Methods of Mathematical Physics* (English translation), Vol. 1, New York: Interscience (1953), reprinting, Wiley (1989), chapter 6, section 3.

We illustrate with the square-wave function

$$f(x) = \begin{cases} \dfrac{h}{2}, & 0 < x < \pi \\ -\dfrac{h}{2}, & -\pi < x < 0. \end{cases} \tag{5.23}$$

The functions $\cos nx$ and $\sin nx$ are orthogonal on the expansion interval (with unit weight in the scalar product), and the expansion of $f(x)$ takes the form

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx).$$

Because $f(x)$ is an odd function of x, all the a_n vanish, and we only need to compute

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt \, dt.$$

The factor $1/\pi$ preceding the integral arises because the expansion functions are not normalized.

Upon substitution of $\pm h/2$ for $f(t)$, we find

$$b_n = \frac{h}{n\pi}(1 - \cos n\pi) = \begin{cases} 0, & n \text{ even}, \\ \dfrac{2h}{n\pi}, & n \text{ odd}. \end{cases}$$

Thus, the expansion of the square wave is

$$f(x) = \frac{2h}{\pi} \sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1}. \tag{5.24}$$

To give an idea of the rate at which the series in Eq. (5.24) converges, some of its partial sums are plotted in Fig. 5.1.

■

Expansions of Dirac Delta Function

Orthogonal expansions provide opportunities to develop additional representations of the Dirac delta function. In fact, such a representation can be built from any complete set of functions $\varphi_n(x)$. For simplicity we assume the φ_n to be orthonormal with unit weight on the interval (a, b), and consider the expansion

$$\delta(x - t) = \sum_{n=0}^{\infty} c_n(t) \, \varphi_n(x), \tag{5.25}$$

where, as indicated, the coefficients must be functions of t. From the rule for determining the coefficients, we have, for t also in the interval (a, b),

$$c_n(t) = \int_a^b \varphi_n^*(x) \, \delta(x - t) \, dx = \varphi_n^*(t), \tag{5.26}$$

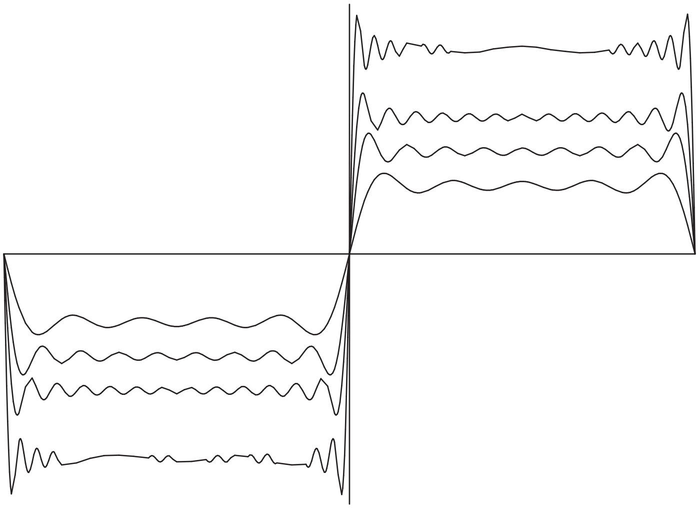

FIGURE 5.1 Expansion of square wave. Computed using Eq. (5.24) with summation terminated after $n = 4$, 8, 12, and 20. Curves are at different vertical scales to enhance visibility.

where the evaluation has used the defining property of the delta function. Substituting this result back into Eq. (5.25), we have

$$\delta(x-t) = \sum_{n=0}^{\infty} \varphi_n^*(t)\,\varphi_n(x). \tag{5.27}$$

This result is clearly not uniformly convergent at $x = t$. However, remember that it is not to be used by itself, but has meaning only when it appears as part of an integrand. Note also that Eq. (5.27) is only valid when x and t are within the range (a, b).

Equation (5.27) is called the **closure** relation for the Dirac delta function (with respect to the φ_n) and obviously depends on the completeness of the φ set. If we apply Eq. (5.27) to an arbitrary function $F(t)$ that we assume to have the expansion $F(t) = \sum_p c_p\,\varphi_p(t)$, we have

$$\begin{aligned}\int_a^b F(t)\,\delta(x-t)\,dt &= \int_a^b dt \sum_{p=0}^{\infty} c_p\,\varphi_p(t) \sum_{n=0}^{\infty} \varphi_n^*(t)\,\varphi_n(x)\\ &= \sum_{p=0}^{\infty} c_p\,\varphi_p(x) = F(x),\end{aligned} \tag{5.28}$$

which is the expected result. However, if we replace the integration limits (a, b) by (t_1, t_2) such that $a \le t_1 < t_2 \le b$, we get a more general result that reflects the fact that our

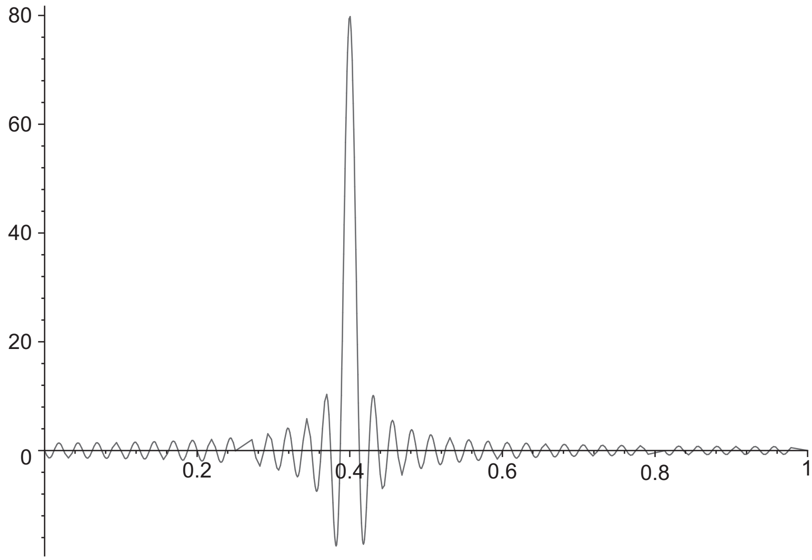

FIGURE 5.2 Approximation at $N = 80$ to $\delta(t - x)$, Eq. (5.30), for $t = 0.4$.

representation of $\delta(x - t)$ is negligible except when $x \approx t$:

$$\int_{t_1}^{t_2} F(t)\,\delta(x-t)\,dt = \begin{cases} F(x), & t_1 < x < t_2, \\ 0, & x < t_1 \text{ or } x > t_2. \end{cases} \tag{5.29}$$

Example 5.1.7 Delta Function Representation

To illustrate an expansion of the Dirac delta function in an orthonormal basis, take $\varphi_n(x) = \sqrt{2}\,\sin n\pi x$, which are orthonormal and complete on $x = (0, 1)$ for $n = 1, 2, \ldots$. Then the Dirac delta function has representation, valid for $0 < x < 1$, $0 < t < 1$,

$$\delta(x - t) = \lim_{N\to\infty} \sum_{n=1}^{N} 2 \sin n\pi t \sin n\pi x. \tag{5.30}$$

Plotting this with $N = 80$ for $t = 0.4$ and $0 < x < 1$ gives the result shown in Fig. 5.2. ■

Dirac Notation

Much of what we have discussed can be brought to a form that promotes clarity and suggests possibilities for additional analysis by using a notational device invented by P. A. M. Dirac. Dirac suggested that instead of just writing a function f, it be written enclosed in the right half of an angle-bracket pair, which he named a **ket**. Thus $f \to |f\rangle$, $\varphi_i \to |\varphi_i\rangle$, etc. Then he suggested that the complex conjugates of functions be enclosed in left half-brackets, which he named **bras**. An example of a bra is $\varphi_i^* \to \langle\varphi_i|$. Finally, he suggested that when the sequence (bra followed by ket = bra+ket ∼ bracket) is encountered, the pair should be interpreted as a scalar product (with the dropping of one of the two adjacent vertical lines). As an initial example of the use of this notation, take Eq. (5.12), which we now write as

$$|f\rangle = \sum_j a_j |\varphi_j\rangle = \sum_j |\varphi_j\rangle\langle\varphi_j|f\rangle = \left(\sum_j |\varphi_j\rangle\langle\varphi_j|\right)|f\rangle. \tag{5.31}$$

This notational rearrangement shows that we can view the expansion in the φ basis as the insertion of a set of basis members in a way which, in sum, has no effect. If the sum is over a complete set of φ_j, the ket-bra sum in Eq. (5.31) will have no net effect when inserted before any ket in the space, and therefore we can view the sum as a **resolution of the identity**. To emphasize this, we write

$$1 = \sum_j |\varphi_j\rangle\langle\varphi_j|. \tag{5.32}$$

Many expressions involving expansions in orthonormal sets can be derived by the insertion of resolutions of the identity.

Dirac notation can also be applied to expressions involving vectors and matrices, where it illuminates the parallelism between physical vector spaces and the function spaces here under study. If $\mathbf{a}$ and $\mathbf{b}$ are column vectors and M is a matrix, then we can write $|\mathbf{b}\rangle$ as a synonym for $\mathbf{b}$, we can write $\langle\mathbf{a}|$ to mean $\mathbf{a}^\dagger$, and then $\langle\mathbf{a}|\mathbf{b}\rangle$ is interpreted as equivalent to $\mathbf{a}^\dagger\mathbf{b}$, which (when the vectors are real) is matrix notation for the (scalar) dot product $\mathbf{a}\cdot\mathbf{b}$. Other examples are expressions such as

$$\mathbf{a} = \mathsf{M}\mathbf{b} \leftrightarrow |\mathbf{a}\rangle = |\mathsf{M}\mathbf{b}\rangle = \mathsf{M}|\mathbf{b}\rangle \quad \text{or} \quad \mathbf{a}^\dagger\mathsf{M}\mathbf{b} = (\mathsf{M}^\dagger\mathbf{a})^\dagger\mathbf{b} \leftrightarrow \langle\mathbf{a}|\mathsf{M}\mathbf{b}\rangle = \langle\mathsf{M}^\dagger\mathbf{a}|\mathbf{b}\rangle.$$

Exercises

5.1.1 A function $f(x)$ is expanded in a series of orthonormal functions

$$f(x) = \sum_{n=0}^{\infty} a_n\varphi_n(x).$$

Show that the series expansion is unique for a given set of $\varphi_n(x)$. The functions $\varphi_n(x)$ are being taken here as the **basis** vectors in an infinite-dimensional Hilbert space.

5.1.2 A function $f(x)$ is represented by a finite set of basis functions $\varphi_i(x)$,

$$f(x) = \sum_{i=1}^{N} c_i\varphi_i(x).$$

Show that the components c_i are unique, that no different set c_i' exists.

Note. Your basis functions are automatically linearly independent. They are not necessarily orthogonal.

5.1.3 A function $f(x)$ is approximated by a power series $\sum_{i=0}^{n-1} c_i x^i$ over the interval [0, 1]. Show that minimizing the mean square error leads to a set of linear equations

$$\mathsf{A}\mathbf{c} = \mathbf{b},$$

where

$$A_{ij} = \int_0^1 x^{i+j}\,dx = \frac{1}{i+j+1}, \quad i, j = 0, 1, 2, \ldots, n-1$$

and

$$b_i = \int_0^1 x^i f(x)\,dx, \quad i = 0, 1, 2, \ldots, n-1.$$

Note. The A_{ij} are the elements of the Hilbert matrix of order n. The determinant of this Hilbert matrix is a rapidly decreasing function of n. For $n = 5$, $\det \mathsf{A} = 3.7 \times 10^{-12}$ and the set of equations $\mathsf{A}c = \mathbf{b}$ is becoming ill-conditioned and unstable.

5.1.4 In place of the expansion of a function $F(x)$ given by

$$F(x) = \sum_{n=0}^{\infty} a_n \varphi_n(x),$$

with

$$a_n = \int_a^b F(x)\varphi_n(x) w(x)\,dx,$$

take the **finite** series approximation

$$F(x) \approx \sum_{n=0}^{m} c_n \varphi_n(x).$$

Show that the mean square error

$$\int_a^b \left[F(x) - \sum_{n=0}^{m} c_n \varphi_n(x) \right]^2 w(x)\,dx$$

is minimized by taking $c_n = a_n$.

Note. The values of the coefficients are independent of the number of terms in the finite series. This independence is a consequence of orthogonality and would not hold for a least-squares fit using powers of x.

5.1.5 From Example 5.1.6,

$$f(x) = \left\{ \begin{array}{ll} \dfrac{h}{2}, & 0 < x < \pi \\ -\dfrac{h}{2}, & -\pi < x < 0 \end{array} \right\} = \frac{2h}{\pi} \sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1}.$$

(a) Show that

$$\int_{-\pi}^{\pi} \left[f(x) \right]^2 dx = \frac{\pi}{2} h^2 = \frac{4h^2}{\pi} \sum_{n=0}^{\infty} (2n+1)^{-2}.$$

For a finite upper limit this would be Bessel's inequality. For the upper limit ∞, this is Parseval's identity.

(b) Verify that

$$\frac{\pi}{2} h^2 = \frac{4h^2}{\pi} \sum_{n=0}^{\infty} (2n+1)^{-2}$$

by evaluating the series.

Hint. The series can be expressed in terms of the Riemann zeta function $\zeta(2) = \pi^2/6$.

5.1.6 Derive the Schwarz inequality from the identity

$$\left[\int_a^b f(x) g(x)\, dx \right]^2 = \int_a^b \left[f(x) \right]^2 dx \int_a^b [\left[g(x) \right]^2 dx$$

$$- \frac{1}{2} \int_a^b dx \int_a^b dy \left[f(x)g(y) - f(y)g(x) \right]^2.$$

5.1.7 Starting from $I = \left\langle f - \sum_i a_i\, \varphi_i \middle| f - \sum_j a_j\, \varphi_j \right\rangle \ge 0$,

derive Bessel's inequality, $\langle f | f \rangle \ge \sum_n |a_n|^2$.

5.1.8 Expand the function $\sin \pi x$ in a series of functions φ_i that are orthogonal (but not normalized) on the range $0 \le x \le 1$ when the scalar product has definition

$$\langle f | g \rangle = \int_0^1 f^*(x) g(x)\, dx.$$

Keep the first four terms of the expansion. The first four φ_i are:

$$\varphi_0 = 1, \quad \varphi_1 = 2x - 1, \quad \varphi_2 = 6x^2 - 6x + 1, \quad \varphi_3 = 20x^3 - 30x^2 + 12x - 1.$$

Note. The integrals that are needed are the subject of Example 1.10.5.

5.1.9 Expand the function e^{-x} in Laguerre polynomials $L_n(x)$, which are orthonormal on the range $0 \le x < \infty$ with scalar product

$$\langle f | g \rangle = \int_0^{\infty} f^*(x) g(x) e^{-x}\, dx.$$

Keep the first four terms of the expansion. The first four $L_n(x)$ are

$$L_0 = 1, \quad L_1 = 1 - x, \quad L_2 = \frac{2 - 4x + x^2}{2}, \quad L_3 = \frac{6 - 18x + 9x^2 - x^3}{6}.$$

5.1.10 The explicit form of a function f is not known, but the coefficients a_n of its expansion in the orthonormal set φ_n are available. Assuming that the φ_n and the members of another orthonormal set, χ_n, are available, use Dirac notation to obtain a formula for the coefficients for the expansion of f in the χ_n set.

5.1.11 Using conventional vector notation, evaluate $\sum_j |\hat{\mathbf{e}}_j\rangle\langle\hat{\mathbf{e}}_j|\mathbf{a}\rangle$, where $\mathbf{a}$ is an arbitrary vector in the space spanned by the $\hat{\mathbf{e}}_j$.

5.1.12 Letting $\mathbf{a} = a_1\hat{\mathbf{e}}_1 + a_2\hat{\mathbf{e}}_2$ and $\mathbf{b} = b_1\hat{\mathbf{e}}_1 + b_2\hat{\mathbf{e}}_2$ be vectors in $\mathbb{R}^2$, for what values of k, if any, is

$$\langle\mathbf{a}|\mathbf{b}\rangle = a_1b_1 - a_1b_2 - a_2b_1 + ka_2b_2$$

a valid definition of a scalar product?

5.2 Gram-Schmidt Orthogonalization

Crucial to carrying out the expansions and transformations under discussion is the availability of useful orthonormal sets of functions. We therefore proceed to the description of a process whereby a set of functions that is neither orthogonal or normalized can be used to construct an orthonormal set that spans the same function space. There are many ways to accomplish this task. We present here the method called the **Gram-Schmidt** orthogonalization process.

The Gram-Schmidt process assumes the availability of a set of functions χ_μ and an appropriately defined scalar product $\langle f|g\rangle$. We orthonormalize **sequentially** to form the orthonormal functions φ_ν, meaning we make the first orthonormal function, φ_0, from χ_0, the next, φ_1, from χ_0 and χ_1, etc. If, for example, the χ_μ are powers x^μ, the orthonormal function φ_ν will be a polynomial of degree ν in x. Because the Gram-Schmidt process is often applied to powers, we have chosen to number both the χ and the φ sets starting from zero (rather than 1).

Thus, our first orthonormal function will simply be a normalized version of χ_0. Specifically,

$$\varphi_0 = \frac{\chi_0}{\langle\chi_0|\chi_0\rangle^{1/2}}. \tag{5.33}$$

To check that Eq. (5.33) is correct, we form

$$\langle\varphi_0|\varphi_0\rangle = \left\langle \frac{\chi_0}{\langle\chi_0|\chi_0\rangle^{1/2}} \middle| \frac{\chi_0}{\langle\chi_0|\chi_0\rangle^{1/2}} \right\rangle = 1.$$

Next, starting from φ_0 and χ_1, we form a function that is orthogonal to φ_0. We use φ_0 rather than χ_0 to be consistent with what we will do in later steps of the process. Thus, we write

$$\psi_1 = \chi_1 - a_{1,0}\varphi_0. \tag{5.34}$$

What we are doing here is the removal from χ_1 of its projection onto φ_0, leaving a remainder that will be orthogonal to φ_0. Remembering that φ_0 is normalized (of "unit length"), that projection is identified as $\langle\varphi_0|\chi_1\rangle\varphi_0$, so that

$$a_{1,0} = \langle\varphi_0|\chi_1\rangle. \tag{5.35}$$

In case Eq. (5.35) is not intuitively obvious, we can confirm it by writing the requirement that ψ_1 be orthogonal to φ_0:

$$\langle\varphi_0|\psi_1\rangle = \left\langle\varphi_0\middle|\left(\chi_1 - a_{1,0}\varphi_0\right)\right\rangle = \langle\varphi_0|\chi_1\rangle - a_{1,0}\langle\varphi_0|\varphi_0\rangle = 0,$$

which, because φ_0 is normalized, reduces to Eq. (5.35). The function ψ_1 is not in general normalized. To normalize it and thereby obtain φ_1, we form

$$\varphi_1 = \frac{\psi_1}{\langle\psi_1|\psi_1\rangle^{1/2}}. \tag{5.36}$$

To continue further, we need to make, from φ_0, φ_1, and χ_2, a function that is orthogonal to both φ_0 and φ_1. It will have the form

$$\psi_2 = \chi_2 - a_{0,2}\varphi_0 - a_{1,2}\varphi_1. \tag{5.37}$$

The last two terms of Eq. (5.37), respectively, remove from χ_2 its projections on φ_0 and φ_1; these projections are independent because φ_0 and φ_1 are orthogonal. Thus, either from our knowledge of projections or by setting to zero the scalar products $\langle\varphi_i|\psi_2\rangle$ ($i = 0$ and 1), we establish

$$a_{0,2} = \langle\varphi_0|\chi_2\rangle, \quad a_{1,2} = \langle\varphi_1|\chi_2\rangle. \tag{5.38}$$

Finally, we make $\varphi_2 = \psi_2/\langle\psi_2|\psi_2\rangle^{1/2}$.

The generalization for which the above is the first few terms is that, given the prior formation of φ_i, $i = 0, \ldots, n-1$, the orthonormal function φ_n is obtained from χ_n by the following two steps:

$$\psi_n = \chi_n - \sum_{\mu=0}^{n-1}\langle\varphi_\mu|\chi_n\rangle\varphi_\mu,$$

$$\varphi_n = \frac{\psi_n}{\langle\psi_n|\psi_n\rangle^{1/2}}. \tag{5.39}$$

Reviewing the above process, we note that different results would have been obtained if we used the same set of χ_i, but simply took them in a different order. For example, if we had started with χ_3, one of our orthonormal functions would have been a multiple of χ_3, while the set we constructed yielded φ_3 as a linear combination of χ_μ, $\mu = 0, 1, 2, 3$.

Example 5.2.1 Legendre Polynomials

Let us form an orthonormal set, taking the χ_μ as x^μ, and making the definition

$$\langle f|g\rangle = \int_{-1}^{1} f^*(x)g(x)dx. \tag{5.40}$$

This scalar product definition will cause the members of our set to be orthogonal, with unit weight, on the range $(-1, 1)$. Moreover, since the χ_μ are real, the complex conjugate asterisk has no operational significance here.

The first orthonormal function, φ_0, is

$$\varphi_0(x) = \frac{1}{\langle 1|1\rangle^{1/2}} = \frac{1}{\left[\int_{-1}^{1} dx\right]^{1/2}} = \frac{1}{\sqrt{2}}.$$

To obtain φ_1, we first obtain ψ_1 by evaluating

$$\psi_1(x) = x - \langle\varphi_0|x\rangle\varphi_0(x) = x,$$

where the scalar product vanishes because φ_0 is an even function of x, whereas x is odd, and the range of integration is even. We then find

$$\varphi_1(x) = \frac{x}{\left[\int_{-1}^{1} x^2 dx\right]^{1/2}} = \sqrt{\frac{3}{2}}\, x.$$

The next step is less trivial. We form

$$\psi_2(x) = x^2 - \langle\varphi_0|x^2\rangle\varphi_0(x) - \langle\varphi_1|x^2\rangle\varphi_1(x) = x^2 - \left\langle\frac{1}{\sqrt{2}}\middle|x^2\right\rangle\left(\frac{1}{\sqrt{2}}\right) = x^2 - \frac{1}{3},$$

where we have used symmetry to set $\langle\varphi_1|x^2\rangle$ to zero and evaluated the scalar product

$$\left\langle\frac{1}{\sqrt{2}}\middle|x^2\right\rangle = \frac{1}{\sqrt{2}}\int_{-1}^{1} x^2 dx = \frac{\sqrt{2}}{3}.$$

Then,

$$\varphi_2(x) = \frac{x^2 - \frac{1}{3}}{\left[\int_{-1}^{1} \left(x^2 - \frac{1}{3}\right)^2 dx\right]^{1/2}} = \sqrt{\frac{5}{2}}\left(\frac{3}{2}x^2 - \frac{1}{2}\right).$$

Continuation to one more orthonormal function yields

$$\varphi_3(x) = \sqrt{\frac{7}{2}}\left(\frac{5}{2}x^3 - \frac{3}{2}x\right).$$

Reference to Chapter 15 will show that

$$\varphi_n(x) = \sqrt{\frac{2n+1}{2}}\, P_n(x), \tag{5.41}$$

where $P_n(x)$ is the nth degree Legendre polynomial. Our Gram-Schmidt process provides a possible but very cumbersome method of generating the Legendre polynomials; other, more efficient approaches exist. ■

Table 5.1 Orthogonal Polynomials Generated by Gram-Schmidt Orthogonalization of $u_n(x) = x^n$, $n = 0,\ 1,\ 2, \ldots$.

Polynomials	Scalar Products	Table
Legendre	$\int_{-1}^{1} P_n(x)P_m(x)dx = 2\delta_{mn}/(2n+1)$	Table 15.1
Shifted Legendre	$\int_{0}^{1} P_n^*(x)P_m^*(x)dx = \delta_{mn}/(2n+1)$	Table 15.2
Chebyshev I	$\int_{-1}^{1} T_n(x)T_m(x)\left(1-x^2\right)^{-1/2}dx = \delta_{mn}\pi/(2-\delta_{n0})$	Table 18.4
Shifted Chebyshev I	$\int_{0}^{1} T_n^*(x)T_m^*(x)[x(1-x)]^{-1/2}dx = \delta_{mn}\pi/(2-\delta_{n0})$	Table 18.5
Chebyshev II	$\int_{-1}^{1} U_n(x)U_m(x)\left(1-x^2\right)^{1/2}dx = \delta_{mn}\pi/2$	Table 18.4
Laguerre	$\int_{0}^{\infty} L_n(x)L_m(x)e^{-x}dx = \delta_{mn}$	Table 18.2
Associated Laguerre	$\int_{0}^{\infty} L_n^k(x)L_m^k(x)e^{-x}dx = \delta_{mn}(n+k)!/n!$	Table 18.3
Hermite	$\int_{-\infty}^{\infty} H_n(x)H_m(x)e^{-x^2}dx = 2^n\delta_{mn}\pi^{1/2}n!$	Table 18.1

The intervals, weights, and conventional normalization can be deduced from the forms of the scalar products. Tables of explicit formulas for the first few polynomials of each type are included in the indicated tables appearing in Chapters 15 and 18 of this book.

The Legendre polynomials are, except for sign and scale, uniquely defined by the Gram-Schmidt process, the use of successive powers of x, and the definition adopted for the scalar product. By changing the scalar product definition (different weight or range), we can generate other useful sets of orthogonal polynomials. A number of these are presented in Table 5.1. For various reasons most of these polynomial sets are not normalized to unity. The scalar product formulas in the table give the conventional normalizations, and are those of the explicit formulas referenced in the table.

Orthonormalizing Physical Vectors

The Gram-Schmidt process also works for ordinary vectors that are simply given by their components, it being understood that the scalar product is just the ordinary dot product.

Example 5.2.2 ORTHONORMALIZING A 2-D MANIFOLD

A 2-D manifold (subspace) in 3-D space is defined by the two vectors $\mathbf{a}_1 = \hat{\mathbf{e}}_1 + \hat{\mathbf{e}}_2 - 2\hat{\mathbf{e}}_3$ and $\mathbf{a}_2 = \hat{\mathbf{e}}_1 + 2\hat{\mathbf{e}}_2 - 3\hat{\mathbf{e}}_3$. In Dirac notation, these vectors (written as column matrices) are

$$|\mathbf{a}_1\rangle = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}, \quad |\mathbf{a}_2\rangle = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}.$$

Our task is to span this manifold with an orthonormal basis.

We proceed exactly as for functions: Our first orthonormal basis vector, which we call $\mathbf{b}_1$, will be a normalized version of $\mathbf{a}_1$, and therefore formed as

$$|\mathbf{b}_1\rangle = \frac{\mathbf{a}_1}{\langle \mathbf{a}_1|\mathbf{a}_1\rangle^{1/2}} = \frac{1}{6^{1/2}}\,|\mathbf{a}_1\rangle = \frac{1}{6^{1/2}}\begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}.$$

An unnormalized version of a second orthonormal function will have the form

$$|\mathbf{b}_2'\rangle = |\mathbf{a}_2\rangle - \langle \mathbf{b}_1|\mathbf{a}_2\rangle|\mathbf{b}_1\rangle = |\mathbf{a}_2\rangle - \frac{9}{6^{1/2}}\,|\mathbf{b}_1\rangle = \begin{pmatrix} -1/2 \\ 1/2 \\ 0 \end{pmatrix}.$$

Normalizing, we reach

$$|\mathbf{b}_2\rangle = \frac{\mathbf{b}_2'}{\langle \mathbf{b}_2'|\mathbf{b}_2'\rangle^{1/2}} = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}.$$

■

Exercises

For the Gram-Schmidt constructions in Exercises 5.2.1 through 5.2.6, use a scalar product of the form given in Eq. (5.7) with the specified interval and weight.

5.2.1 Following the Gram-Schmidt procedure, construct a set of polynomials $P_n^*(x)$ orthogonal (unit weighting factor) over the range [0, 1] from the set $[1,\ x,\ x^2,\ \ldots]$. Scale so that $P_n^*(1) = 1$.

ANS.
$$\begin{aligned} P_n^*(x) &= 1, \\ P_1^*(x) &= 2x - 1, \\ P_2^*(x) &= 6x^2 - 6x + 1, \\ P_3^*(x) &= 20x^3 - 30x^2 + 12x - 1. \end{aligned}$$

These are the first four **shifted** Legendre polynomials.

Note. The "*" is the standard notation for "shifted": [0, 1] instead of [−1, 1]. It does **not** mean complex conjugate.

5.2.2 Apply the Gram-Schmidt procedure to form the first three Laguerre polynomials:

$$u_n(x) = x^n, \quad n = 0, 1, 2, \ldots, \quad 0 \le x < \infty, \quad w(x) = e^{-x}.$$

The conventional normalization is

$$\int_0^\infty L_m(x)L_n(x)e^{-x}\,dx = \delta_{mn}.$$

$$\textit{ANS.}\quad L_0 = 1,\quad L_1 = (1-x),\quad L_2 = \frac{2-4x+x^2}{2}.$$

5.2.3 You are given

(a) a set of functions $u_n(x) = x^n$, $n = 0, 1, 2, \ldots,$

(b) an interval $(0, \infty)$,

(c) a weighting function $w(x) = xe^{-x}$. Use the Gram-Schmidt procedure to construct the first **three orthonormal** functions from the set $u_n(x)$ for this interval and this weighting function.

$$\textit{ANS.}\quad \varphi_0(x) = 1,\quad \varphi_1(x) = (x-2)/\sqrt{2},\quad \varphi_2(x) = (x^2-6x+6)/2\sqrt{3}.$$

5.2.4 Using the Gram-Schmidt orthogonalization procedure, construct the lowest three Hermite polynomials:

$$u_n(x) = x^n,\quad n = 0, 1, 2, \ldots,\quad -\infty < x < \infty,\quad w(x) = e^{-x^2}.$$

For this set of polynomials the usual normalization is

$$\int_{-\infty}^{\infty} H_m(x)H_n(x)w(x)\,dx = \delta_{mn}2^m m!\,\pi^{1/2}.$$

$$\textit{ANS.}\quad H_0 = 1,\quad H_1 = 2x,\quad H_2 = 4x^2 - 2.$$

5.2.5 Use the Gram-Schmidt orthogonalization scheme to construct the first three Chebyshev polynomials (type I):

$$u_n(x) = x^n,\quad n = 0, 1, 2, \ldots,\quad -1 \le x \le 1,\quad w(x) = (1-x^2)^{-1/2}.$$

Take the normalization

$$\int_{-1}^{1} T_m(x)T_n(x)w(x)dx = \delta_{mn}\begin{cases}\pi, & m = n = 0,\\ \dfrac{\pi}{2}, & m = n \ge 1.\end{cases}$$

Hint. The needed integrals are given in Exercise 13.3.2.

$$\textit{ANS.}\quad T_0 = 1,\quad T_1 = x,\quad T_2 = 2x^2 - 1,\quad (T_3 = 4x^3 - 3x).$$

5.2.6 Use the Gram-Schmidt orthogonalization scheme to construct the first three Chebyshev polynomials (type II):

$$u_n(x) = x^n,\quad n = 0, 1, 2, \ldots,\quad -1 \le x \le 1,\quad w(x) = (1-x^2)^{+1/2}.$$

Take the normalization to be

$$\int_{-1}^{1} U_m(x)U_n(x)w(x)\,dx = \delta_{mn}\frac{\pi}{2}.$$

Hint.

$$\int_{-1}^{1} (1-x^2)^{1/2} x^{2n}\,dx = \frac{\pi}{2} \times \frac{1\cdot 3\cdot 5\cdots(2n-1)}{4\cdot 6\cdot 8\cdots(2n+2)}, \quad n = 1, 2, 3, \ldots$$
$$= \frac{\pi}{2}, \quad n = 0.$$

ANS. $U_0 = 1, \quad U_1 = 2x, \quad U_2 = 4x^2 - 1.$

5.2.7 As a modification of Exercise 5.2.5, apply the Gram-Schmidt orthogonalization procedure to the set $u_n(x) = x^n$, $n = 0, 1, 2, \ldots$, $0 \le x < \infty$. Take $w(x)$ to be $\exp(-x^2)$. Find the first two nonvanishing polynomials. Normalize so that the coefficient of the highest power of x is unity. In Exercise 5.2.5, the interval $(-\infty, \infty)$ led to the Hermite polynomials. The functions found here are certainly not the Hermite polynomials.

ANS. $\varphi_0 = 1, \quad \varphi_1 = x - \pi^{-1/2}.$

5.2.8 Form a set of three orthonormal vectors by the Gram-Schmidt process using these input vectors in the order given:

$$\mathbf{c}_1 = \begin{pmatrix}1\\1\\1\end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix}1\\1\\2\end{pmatrix}, \quad \mathbf{c}_3 = \begin{pmatrix}1\\0\\2\end{pmatrix}.$$

5.3 OPERATORS

An operator is a mapping between functions in its **domain** (those to which it can be applied) and functions in its **range** (those it can produce). While the domain and the range need not be in the same space, our concern here is for operators whose domain and range are both in all or part of the same Hilbert space. To make this discussion more concrete, here are a few examples of operators:

- Multiplication by 2: Converts f into $2f$;
- For a space containing algebraic functions of a variable x, d/dx: Converts $f(x)$ into df/dx;
- An integral operator A defined by $Af(x) = \int G(x, x') f(x')\,dx'$: A special case of this is a projection operator $|\varphi_i\rangle\langle\varphi_i|$, which converts f into $\langle\varphi_i|f\rangle\varphi_i$.

In addition to the abovementioned restriction on domain and range, we also for our present purposes restrict attention to operators that are **linear**, meaning that if A and B are linear operators, f and g functions, and k a constant, then

$$(A + B)f = Af + Bf, \quad A(f + g) = Af + Ag, \quad Ak = kA.$$

For both electromagnetic theory and quantum mechanics, an important class of operators are **differential operators**, those that include differentiation of the functions to which they are applied. These operators arise when differential equations are written in operator form;

for example, the operator

$$\mathcal{L}(x) = \left(1 - x^2\right)\frac{d^2}{dx^2} - 2x\frac{d}{dx}$$

enables us to write Legendre's differential equation,

$$\left(1 - x^2\right)\frac{d^2 y}{dx^2} - 2x\frac{dy}{dx} + \lambda y = 0,$$

in the form $\mathcal{L}(y)y = -\lambda y$. When no confusion thereby results, this can be shortened to $\mathcal{L}y = -\lambda y$.

Commutation of Operators

Because differential operators act on the function(s) to their right, they do not necessarily commute with other operators containing the same independent variable. This fact makes it useful to consider the **commutator** of operators A and B,

$$[A, B] = AB - BA. \tag{5.42}$$

We can often reduce $AB - BA$ to a simpler operator expression. When we write an operator equation, its meaning is that the operator on the left-hand side of the equation produces the same effect on every function in its domain as is produced by the operator on the right-hand side. Let's illustrate this point by evaluating the commutator $[x, p]$, where $p = -i\, d/dx$. The imaginary unit i and the name p appear because this operator is that corresponding in quantum mechanics to momentum (in a system of units such that $\hbar = h/2\pi = 1$). The operator x stands for multiplication by x.

To carry out the evaluation, we apply $[x, p]$ to an arbitrary function $f(x)$. Inserting the explicit form of p, we have

$$[x, p]f(x) = (xp - px)f(x) = -ix\frac{df(x)}{dx} - \left(-i\frac{d}{dx}\right)\Big(x\, f(x)\Big)$$
$$= -ixf'(x) + i\Big(f(x) + xf'(x)\Big) = i\, f(x),$$

indicating that

$$[x, p] = i. \tag{5.43}$$

As indicated before, this means $[x, p]f(x) = i\, f(x)$ for all f.

We can carry out various algebraic manipulations on commutators. In general, if A, B, C are operators and k is a constant,

$$[A, B] = -[B, A], \qquad [A, B + C] = [A, B] + [A, C], \qquad k[A, B] = [kA, B] = [A, kB]. \tag{5.44}$$

Example 5.3.1 Operator Manipulation

Given $[x, p]$, we can simplify the commutator $[x, p^2]$. We write, being careful about the operator ordering and using Eq. (5.43),

$$[x, p^2] = xp^2 - pxp + pxp - p^2x = [x, p]p + p[x, p] = 2i\, p, \tag{5.45}$$

a result also obtainable from

$$x\left(-\frac{d^2}{dx^2}\right) f(x) - \left(-\frac{d^2}{dx^2}\right) x f(x) = 2f'(x) = 2i\left(-i\frac{d}{dx}\right) f(x).$$

However, note that Eq. (5.45) follows solely from the validity of Eq. (5.43), and will apply to any quantities x and p that satisfy that commutation relation, whether or not we are operating with ordinary functions and their derivatives. Put another way, if x and p are operators in some abstract Hilbert space and all we know about them is Eq. (5.43), we may still conclude that Eq. (5.45) is also valid. ■

Identity, Inverse, Adjoint

An operator that is generally available is the **identity operator**, namely one that leaves functions unchanged. Depending on the context, this operator will be denoted either I or simply **1**. Some, but not all operators will have an inverse, namely an operator that will "undo" its effect. Letting A^{-1} denote the inverse of A, if A^{-1} exists, it will have the property

$$A^{-1}A = AA^{-1} = 1. \tag{5.46}$$

Associated with many operators will be another operator, called its **adjoint** and denoted $A^\dagger$, which will be such that for all functions f and g in the Hilbert space,

$$\langle f|Ag\rangle = \langle A^\dagger f|g\rangle. \tag{5.47}$$

Thus, we see that $A^\dagger$ is an operator that, applied to the left member of **any** scalar product, produces the same result as is obtained if A is applied to the right member of the same scalar product. Equation (5.47) is, in essence, the defining equation for $A^\dagger$.

Depending on the specific operator A, and the definitions in use of the Hilbert space and the scalar product, $A^\dagger$ may or may not be equal to A. If $A = A^\dagger$, A is referred to as **self-adjoint**, or equivalently, **Hermitian**. If $A^\dagger = -A$, A is called **anti-Hermitian**. This definition is worth emphasis:

$$\text{If} \quad H^\dagger = H, \quad H \text{ is Hermitian.} \tag{5.48}$$

Another situation of frequent occurrence is that the adjoint of an operator is equal to its inverse, in which case the operator is called **unitary**. A unitary operator U is therefore defined by the following statement:

$$\text{If} \quad U^\dagger = U^{-1}, \quad U \text{ is unitary.} \tag{5.49}$$

In the special case that U is both real and unitary, it is called **orthogonal**.

The reader will doubtless note that the nomenclature for operators is similar to that previously introduced for matrices. This is not accidental; we shall shortly develop correspondences between operator and matrix expressions.

Example 5.3.2 FINDING THE ADJOINT

Consider an operator $A = x(d/dx)$ whose domain is the Hilbert space whose members f have a finite value of $\langle f|f\rangle$ when the scalar product has definition

$$\langle f|g\rangle = \int_{-\infty}^{\infty} f^*(x)g(x)\,dx.$$

This space is often referred to as $\mathcal{L}^2$ on $(-\infty, \infty)$. Starting from $\langle f|A\,g\rangle$, we integrate by parts as needed to move the operator out of the right half of the scalar product. Because f and g must vanish at $\pm\infty$, the integrated terms vanish, and we get

$$\langle f|A\,g\rangle = \int_{-\infty}^{\infty} f^* x \frac{dg}{dx}\,dx = \int_{-\infty}^{\infty} \big(xf^*\big)\frac{dg}{dx}\,dx = -\int_{-\infty}^{\infty} \frac{d\big(xf^*\big)}{dx}\,g\,dx$$
$$= \left\langle -\left(\frac{d}{dx}\right)xf \,\middle|\, g\right\rangle.$$

We see from the above that $A^\dagger = -(d/dx)x$, from which we can find $A^\dagger = -A - 1$. This A is clearly neither Hermitian nor unitary (with the specified definition of the scalar product). ■

Example 5.3.3 ADJOINT DEPENDS ON SCALAR PRODUCT

For the Hilbert space and scalar product of Example 5.3.2, an integration by parts easily establishes that an operator $A = -i(d/dx)$ is self-adjoint, i.e., $A^\dagger = A$. But now let's consider the same operator A, but for the $\mathcal{L}^2$ space with $-1 \le x \le 1$ (and with a scalar product of the same form, but with integration limits ±1). In this space, the integrated terms from the integration by parts do not vanish, but we can incorporate them into an operator on the left half of the scalar product by adding delta-function terms:

$$\left\langle f \,\middle|\, -i\frac{d}{dx} \,\middle|\, g\right\rangle = -if^*g\Big|_{-1}^{1} + \int_{-1}^{1}\left(-i\frac{df}{dx}\right)^* g\,dx$$
$$= \int_{-1}^{1}\left(\left[i\delta(x-1) - i\delta(x+1) - i\frac{d}{dx}\right]f(x)\right)^* g(x)\,dx.$$

In this truncated space the operator A is **not** self-adjoint. ■

Basis Expansions of Operators

Because we are dealing only with linear operators, we can write the effect of an operator on an arbitrary function if we know the result of its action on all members of a basis spanning our Hilbert space. In particular, assume that the action of an operator A on member φ_μ of an orthonormal basis has the result, also expanded in that basis,

$$A\varphi_\mu = \sum_\nu a_{\nu\mu}\varphi_\nu. \tag{5.50}$$

Assuming this form for the result of operation with A is not a major restriction; all it says is that the result is in our Hilbert space. Formally, the coefficients $a_{\nu\mu}$ can be obtained by taking scalar products:

$$a_{\nu\mu} = \langle\varphi_\nu|A\varphi_\mu\rangle = \langle\varphi_\nu|A|\varphi_\mu\rangle. \tag{5.51}$$

Following common usage, we have inserted an optional (operationally meaningless) vertical line between A and φ_μ. This notation has the aesthetic effect of separating the operator from the two functions entering the scalar product, and also emphasizes the possibility that instead of evaluating the scalar product as written, we can without changing its value evaluate it using the adjoint of A, as $\langle A^\dagger\varphi_\nu|\varphi_\mu\rangle$.

We now apply Eq. (5.50) to a function ψ whose expansion in the φ basis is

$$\psi = \sum_\mu c_\mu\varphi_\mu, \quad c_\mu = \langle\varphi_\mu|\psi\rangle. \tag{5.52}$$

The result is

$$A\psi = \sum_\mu c_\mu\, A\varphi_\mu = \sum_\mu c_\mu \sum_\nu a_{\nu\mu}\varphi_\nu = \sum_\nu \left(\sum_\mu a_{\nu\mu}c_\mu\right)\varphi_\nu. \tag{5.53}$$

If we think of $A\psi$ as a function χ in our Hilbert space, with expansion

$$\chi = \sum_\nu b_\nu\varphi_\nu, \tag{5.54}$$

we then see from Eq. (5.53) that the coefficients b_ν are related to c_μ and $a_{\nu\mu}$ in a way corresponding to matrix multiplication. To make this more concrete,

- Define $\mathbf{c}$ as a column vector with elements c_i, representing the function ψ,
- Define $\mathbf{b}$ as a column vector with elements b_i, representing the function χ,
- Define A as a matrix with elements a_{ij}, representing the operator A,
- The operator equation $\chi = A\psi$ then corresponds to the matrix equation $\mathbf{b} = \mathsf{A}\mathbf{c}$.

In other words, the expansion of the result of applying A to any function ψ can be computed (by matrix multiplication) from the expansions of A and ψ. In effect, that means that the operator A can be thought of as completely defined by its **matrix elements**, while ψ and $\chi = A\psi$ are completely characterized by their coefficients.

We obtain an interesting expression if we introduce Dirac notation for all the quantities entering Eq. (5.53). We then have, moving the ket representing φ_ν to the left,

$$A\psi = \sum_{\nu\mu} |\varphi_\nu\rangle\langle\varphi_\nu|A|\varphi_\mu\rangle\langle\varphi_\mu|\psi\rangle, \tag{5.55}$$

which leads us to identify A as

$$A = \sum_{\nu\mu} |\varphi_\nu\rangle\langle\varphi_\nu|A|\varphi_\mu\rangle\langle\varphi_\mu|, \tag{5.56}$$

which we note is nothing other than A, multiplied on each side by a resolution of the identity, of the form given in Eq. (5.32).

Another interesting observation results if we reintroduce into Eq. (5.56) the coefficient $a_{\nu\mu}$, bringing us to

$$A = \sum_{\nu\mu} |\varphi_\nu\rangle a_{\nu\mu}\langle\varphi_\mu|. \tag{5.57}$$

Here we have the general form for an operator A, with a specific behavior that is determined entirely by the set of coefficients $a_{\nu\mu}$. The special case $A = 1$ has already been seen to be of the form of Eq. (5.57) with $a_{\nu\mu} = \delta_{\nu\mu}$.

Example 5.3.4 MATRIX ELEMENTS OF AN OPERATOR

Consider the expansion of the operator x in a basis consisting of functions $\varphi_n(x) = C_n H_n(x)e^{-x^2/2}$, $n = 0, 1, \dots,$ where the H_n are Hermite polynomials, with scalar product

$$\langle f|g\rangle = \int_{-\infty}^{\infty} f^*(x)g(x)\,dx.$$

From Table 5.1, we can see that the φ_n are orthogonal and that they will also be normalized if $C_n = (2^n n!\sqrt{\pi})^{-1/2}$. The matrix elements of x, which we denote $x_{\nu\mu}$ and are written collectively as a matrix denoted x, are given by

$$x_{\nu\mu} = \langle\varphi_\nu|x|\varphi_\mu\rangle = C_\nu C_\mu \int_{-\infty}^{\infty} H_\nu(x)\, x\, H_\mu(x)e^{-x^2}\,dx.$$

The integral leading to $x_{\nu\mu}$ can be evaluated in general by using the properties of the Hermite polynomials, but our present purposes are adequately served by a straightforward case-by-case computation. From the table of Hermite polynomials in Table 18.1, we identify

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2, \quad H_3 = 8x^3 - 12x, \quad \dots,$$

and we take note of the integration formula

$$I_n = \int_{-\infty}^{\infty} x^{2n}e^{-x^2}\,dx = \frac{(2n-1)!!\sqrt{\pi}}{2^n}.$$

Making use of the parity (even/odd symmetry) of the H_n and the fact that the matrix x is symmetric, we note that many matrix elements are either zero or equal to others. We illustrate with the explicit computation of one matrix element, x_{12}:

$$x_{12} = C_1 C_2 \int_{-\infty}^{\infty} (2x)x(4x^2-2)e^{-x^2}\,dx = C_1 C_2 \int_{-\infty}^{\infty} \left(8x^4 - 4x^2\right)e^{-x^2}\,dx$$
$$= C_1 C_2 \Big[8I_2 - 4I_1 \Big] = 1.$$

Evaluating other matrix elements, we find that x, the matrix of x, has the form

$$\mathsf{x} = \begin{pmatrix} 0 & \sqrt{2}/2 & 0 & 0 & \cdots \\ \sqrt{2}/2 & 0 & 1 & 0 & \cdots \\ 0 & 1 & 0 & \sqrt{6}/2 & \cdots \\ 0 & 0 & \sqrt{6}/2 & 0 & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \end{pmatrix}. \tag{5.58}$$

■

Basis Expansion of Adjoint

We now look at the adjoint of our operator A as an expansion in the same basis. Our starting point is the definition of the adjoint. For arbitrary functions ψ and χ,

$$\langle\psi|A|\chi\rangle = \langle A^\dagger\psi|\chi\rangle = \langle\chi|A^\dagger|\psi\rangle^*,$$

where we reached the last member of the equation by using the complex conjugation property of the scalar product. This is equivalent to

$$\langle\chi|A^\dagger|\psi\rangle = \langle\psi|A|\chi\rangle^* = \left[\langle\psi|\left(\sum_{\nu\mu}|\varphi_\nu\rangle a_{\nu\mu}\langle\varphi_\mu|\right)|\chi\rangle\right]^*$$
$$= \sum_{\nu\mu}\langle\psi|\varphi_\nu\rangle^* a_{\nu\mu}^* \langle\varphi_\mu|\chi\rangle^*$$
$$= \sum_{\nu\mu}\langle\chi|\varphi_\mu\rangle a_{\nu\mu}^* \langle\varphi_\nu|\psi\rangle, \tag{5.59}$$

where in the last line we have again used the scalar product complex conjugation property and have reordered the factors in the sum.

We are now in a position to note that Eq. (5.59) corresponds to

$$A^\dagger = \sum_{\nu\mu}|\varphi_\nu\rangle a_{\mu\nu}^* \langle\varphi_\mu|. \tag{5.60}$$

In writing Eq. (5.60) we have changed the dummy indices to make the formula as similar as possible to Eq. (5.57). It is important to note the differences: The coefficient $a_{\nu\mu}$ of Eq. (5.57) has been replaced by $a_{\mu\nu}^*$, so we see that the index order has been reversed and the complex conjugate taken. This is the general recipe for forming the basis set expansion of the adjoint of an operator. The relation between the matrix elements of A and of $A^\dagger$ is exactly that which relates a matrix A to its adjoint $\mathsf{A}^\dagger$, showing that the similarity in nomenclature is purposeful. We thus have the important and general result:

- If A is the matrix representing an operator A, then the operator $A^\dagger$, the adjoint of A, is represented by the matrix $\mathsf{A}^\dagger$.

Example 5.3.5 ADJOINT OF SPIN OPERATOR

Consider a spin space spanned by functions we call α and β, with a scalar product completely defined by the equations $\langle\alpha|\alpha\rangle = \langle\beta|\beta\rangle = 1$, $\langle\alpha|\beta\rangle = 0$. An operator B is such that

$$B\,\alpha = 0, \quad B\,\beta = \alpha.$$

Taking all possible linearly independent scalar products, this means that

$$\langle\alpha|B\alpha\rangle = 0, \quad \langle\beta|B\alpha\rangle = 0, \quad \langle\alpha|B\beta\rangle = 1, \quad \langle\beta|B\beta\rangle = 0.$$

It is therefore necessary that

$$\langle B^\dagger\alpha|\alpha\rangle = 0, \quad \langle B^\dagger\beta|\alpha\rangle = 0, \quad \langle B^\dagger\alpha|\beta\rangle = 1, \quad \langle B^\dagger\beta|\beta\rangle = 0,$$

which means that $B^\dagger$ is an operator such that

$$B^\dagger\,\alpha = \beta, \quad B^\dagger\,\beta = 0.$$

The above equations correspond to the matrices

$$\mathsf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathsf{B}^\dagger = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

We see that $\mathsf{B}^\dagger$ is the adjoint of B, as required. ■

Functions of Operators

Our ability to represent operators by matrices also implies that the observations made in Chapter 3 regarding functions of matrices also apply to linear operators. Thus, we have definite meanings for quantities such as $\exp(A)$, $\sin(A)$, or $\cos(A)$, and can also apply to operators various identities involving matrix commutators. Important examples include the Jacobi identity (Exercise 2.2.7), and the Baker-Hausdorff formula, Eq. (2.85).

Exercises

5.3.1 Show (without introducing matrix representations) that the adjoint of the adjoint of an operator restores the original operator, i.e., that $(A^\dagger)^\dagger = A$.

5.3.2 U and V are two arbitrary operators. Without introducing matrix representations of these operators, show that

$$(UV)^\dagger = V^\dagger U^\dagger.$$

Note the resemblance to adjoint matrices.

5.3.3 Consider a Hilbert space spanned by the three functions $\varphi_1 = x_1$, $\varphi_2 = x_2$, $\varphi_3 = x_3$, and a scalar product defined by $\langle x_\nu | x_\mu \rangle = \delta_{\nu\mu}$.

(a) Form the 3×3 matrix of each of the following operators:

$$A_1 = \sum_{i=1}^{3} x_i \left(\frac{\partial}{\partial x_i} \right), \quad A_2 = x_1 \left(\frac{\partial}{\partial x_2} \right) - x_2 \left(\frac{\partial}{\partial x_1} \right).$$

(b) Form the column vector representing $\psi = x_1 - 2x_2 + 3x_3$.

(c) Form the matrix equation corresponding to $\chi = (A_1 - A_2)\psi$ and verify that the matrix equation reproduces the result obtained by direct application of $A_1 - A_2$ to ψ.

5.3.4 (a) Obtain the matrix representation of $A = x(d/dx)$ in a basis of Legendre polynomials, keeping terms through P_3. Use the orthonormal forms of these polynomials as given in 5.2.1 and the scalar product defined there.

(b) Expand x^3 in the orthonormal Legendre polynomial basis.

(c) Verify that Ax^3 is given correctly by its matrix representation.

5.4 SELF-ADJOINT OPERATORS

Operators that are self-adjoint (Hermitian) are of particular importance in quantum mechanics because observable quantities are associated with Hermitian operators. In particular, the average value of an observable A in a quantum mechanical state described by any normalized wave function ψ is given by the **expectation value** of A, defined as

$$\langle A \rangle = \langle \psi | A | \psi \rangle. \tag{5.61}$$

This, of course, only makes sense if it can be assured that $\langle A \rangle$ is real, even if ψ and/or A is complex. Using the fact that A is postulated to be Hermitian, we take the complex conjugate of $\langle A \rangle$:

$$\langle A \rangle^* = \langle \psi | A | \psi \rangle^* = \langle A\psi | \psi \rangle,$$

which reduces to $\langle A \rangle$ because A is self-adjoint.

We have already seen that if A and $A^\dagger$ are expanded in a basis, the matrix $\mathsf{A}^\dagger$ must be the matrix adjoint of the matrix A. This means that the coefficients in its expansion must satisfy

$$a_{\nu\mu} = a^*_{\mu\nu} \quad \text{(coefficients of self-adjoint } A\text{)}. \tag{5.62}$$

Thus, we have the nearly self-evident result: A matrix representing a Hermitian operator is a Hermitian matrix. It is also obvious from Eq. (5.62) that the diagonal elements of a Hermitian matrix (which are expectation values for the basis functions) are real.

We can easily verify from basis expansions that $\langle A\rangle$ must be real. Letting $\mathbf{c}$ be the vector of expansion coefficients of ψ in the basis for which $a_{\nu\mu}$ are the matrix elements of A, then

$$\langle A\rangle = \langle\psi|A|\psi\rangle = \left\langle \sum_\nu c_\nu\varphi_\nu \middle| A \middle| \sum_\mu c_\mu\varphi_\mu \right\rangle = \sum_{\nu\mu} c_\nu^* \langle\varphi_\nu|A|\varphi_\mu\rangle c_\mu$$
$$= \sum_{\nu\mu} c_\nu^* a_{\nu\mu} c_\mu = \mathbf{c}^\dagger \mathsf{A}\mathbf{c},$$

which reduces, as it must, to a scalar. Because A is a self-adjoint matrix, $\mathbf{c}^\dagger\mathsf{A}\mathbf{c}$ is easily seen to be a self-adjoint 1×1 matrix, i.e., a **real** scalar (use the facts that $(\mathsf{BAC})^\dagger = \mathsf{C}^\dagger\mathsf{A}^\dagger\mathsf{B}^\dagger$ and that $\mathsf{A}^\dagger = \mathsf{A}$).

Example 5.4.1 Some Self-Adjoint Operators

Consider the operators x and p introduced earlier, with a scalar product of definition

$$\langle f|g\rangle = \int_{-\infty}^{\infty} f^*(x)g(x)\,dx, \tag{5.63}$$

where our Hilbert space is the set of all functions f for which $\langle f|f\rangle$ exists (i.e., $\langle f|f\rangle$ is finite). This is the $\mathcal{L}^2$ space on the interval $(-\infty,\infty)$. To test whether x is self-adjoint, we compare $\langle f|xg\rangle$ and $\langle xf|g\rangle$. Writing these out as integrals, we consider

$$\int_{-\infty}^{\infty} f^*(x)x\,g(x)dx \quad \text{vs.} \quad \int_{-\infty}^{\infty} [xf(x)]^*\,g(x)dx.$$

Because the order of ordinary functions (including x) can be changed without affecting the value of an integral, and because x is inherently real, these two expressions are equal and x is self-adjoint.

Turning next to $p = -i(d/dx)$, the comparison we must make is

$$\int_{-\infty}^{\infty} f^*(x)\left[-i\frac{d\,g(x)}{dx}\right]dx \quad \text{vs.} \quad \int_{-\infty}^{\infty} \left[-i\frac{df(x)}{dx}\right]^* g(x)dx. \tag{5.64}$$

We can bring these expressions into better correspondence if we integrate the first by parts, differentiating $f^*(x)$ and integrating $dg(x)/dx$. Doing so, the first expression above becomes

$$\int_{-\infty}^{\infty} f^*(x)\left[-i\frac{d\,g(x)}{dx}\right]dx = -if^*(x)g(x)\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty}\left[\frac{df(x)}{dx}\right]^*\Big[-ig(x)\Big]dx.$$

The boundary terms to be evaluated at $\pm\infty$ must vanish because $\langle f|f\rangle$ and $\langle g|g\rangle$ are finite, which assures also (from the Schwarz inequality) that $\langle f|g\rangle$ is finite as well. Upon moving i within the complex conjugate in the remaining integral, we verify agreement with the second expression in Eq. (5.64). Thus both x and p are self-adjoint. Note that if p had not contained the factor i, it would **not** have been self-adjoint, as we obtained a needed sign change when i was moved within the scope of the complex conjugate. ■

Example 5.4.2 Expectation Value of p

Because p, though Hermitian, is also imaginary, consider what happens when we compute its expectation value for a wave function of the form $\psi(x) = e^{i\theta} f(x)$, where $f(x)$ is a real $\mathcal{L}^2$ wave function and θ is a real phase angle. Using the scalar product as defined in Eq. (5.63), and remembering that $p = -i(d/dx)$, we have

$$\langle p\rangle = -i\int_{-\infty}^{\infty} f(x)\frac{df(x)}{dx}\,dx = -\frac{i}{2}\int_{-\infty}^{\infty}\frac{d}{dx}\Big[f(x)\Big]^2 dx$$
$$= -\frac{i}{2}\Big[f(+\infty)^2 - f(-\infty)^2\Big] = 0.$$

As shown, this integral vanishes because $f(x) = 0$ at $\pm\infty$ (this is fortunate because expectation values must be real). This result corresponds to the well-known property that wave functions that describe time-dependent phenomena (nonzero momentum) cannot either be real or real except for a constant (complex) phase factor. ■

The relations between operators and their adjoints provide opportunities for rearrangements of operator expressions that may facilitate their evaluation. Some examples follow.

Example 5.4.3 Operator Expressions

(a) Suppose we wish to evaluate $\langle (x^2 + p^2)\psi|\varphi\rangle$, with ψ of a complicated functional form that might be unpleasant to differentiate (as required to apply p^2), whereas φ is simple. Because x is self-adjoint, so also is x^2:

$$\langle x^2\psi|\varphi\rangle = \langle x\psi|x\varphi\rangle = \langle\psi|x^2\varphi\rangle.$$

The same is true of p^2, so $\langle (x^2 + p^2)\psi|\varphi\rangle = \langle\psi|(x^2 + p^2)\varphi\rangle$.

(b) Look next at $\langle (x + ip)\psi|(x + ip)\psi\rangle$, which is the expression to be evaluated if we want the norm of $(x + ip)\psi$. Note that $x + ip$ is **not** self-adjoint, but has adjoint $x - ip$. Our norm rearranges to

$$\begin{aligned}\langle (x+ip)\psi|(x+ip)\psi\rangle &= \langle\psi|(x-ip)(x+ip)|\psi\rangle\\ &= \langle\psi|x^2 + p^2 + i(xp - px)|\psi\rangle\\ &= \langle\psi|x^2 + p^2 + i(i)|\psi\rangle = \langle\psi|x^2 + p^2 - 1|\psi\rangle.\end{aligned}$$

To reach the last line of the above equation, we recognized the commutator $[x, p] = i$, as established in Eq. (5.43).

(c) Suppose that A and B are self-adjoint. What can we say about the self-adjointness of AB? Consider

$$\langle\psi|AB|\varphi\rangle = \langle A\psi|B|\varphi\rangle = \langle BA\psi|\varphi\rangle.$$

Note that because we moved A to the left first (with no dagger needed because it is self-adjoint), it is part of what the subsequently moved B must operate on. So we see that the adjoint of AB is BA. We conclude that AB is only self-adjoint if A and B commute (so that $BA = AB$). Note that if A and B were not individually self-adjoint, their commutation would not be sufficient to make AB self-adjoint.

■

Exercises

5.4.1 (a) A is a non-Hermitian operator. Show that the operators $A + A^\dagger$ and $i(A - A^\dagger)$ are Hermitian.

(b) Using the preceding result, show that every non-Hermitian operator may be written as a linear combination of two Hermitian operators.

5.4.2 Prove that the product of two Hermitian operators is Hermitian if and only if the two operators commute.

5.4.3 A and B are noncommuting quantum mechanical operators, and C is given by the formula

$$AB - BA = iC.$$

Show that C is Hermitian. Assume that appropriate boundary conditions are satisfied.

5.4.4 The operator $\mathcal{L}$ is Hermitian. Show that $\langle\mathcal{L}^2\rangle \geq 0$, meaning that for all ψ in the space in which $\mathcal{L}$ is defined, $\langle\psi|\mathcal{L}^2|\psi\rangle \geq 0$.

5.4.5 Consider a Hilbert space whose members are functions defined on the surface of the unit sphere, with a scalar product of the form

$$\langle f|g\rangle = \int d\Omega\, f^* g,$$

where $d\Omega$ is the element of solid angle. Note that the total solid angle of the sphere is 4π. We work here with the three functions $\varphi_1 = Cx/r$, $\varphi_2 = Cy/r$, $\varphi_3 = Cz/r$, with C assigned a value that makes the φ_i normalized.

(a) Find C, and show that the φ_i are also mutually orthogonal.

(b) Form the 3×3 matrices of the angular momentum operators

$$L_x = -i\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \quad L_y = -i\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right),$$
$$L_z = -i\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right).$$

(c) Verify that the matrix representations of the components of **L** satisfy the angular momentum commutator $[L_x, L_y] = iL_z$.

5.5 UNITARY OPERATORS

One of the reasons unitary operators are important in physics is that they can be used to describe transformations between orthonormal bases. This property is the generalization to the complex domain of the rotational transformations of ordinary (physical) vectors that we analyzed in Chapter 3.

Unitary Transformations

Suppose we have a function ψ that has been expanded in the orthonormal basis φ:

$$\psi = \sum_\mu c_\mu \varphi_\mu = \left(\sum_\mu |\varphi_\mu\rangle\langle\varphi_\mu| \right) |\psi\rangle. \tag{5.65}$$

We now wish to convert this expansion to a different orthonormal basis, with functions φ'_ν. A possible starting point is to recognize that each of the original basis functions can be expanded in the primed basis. We can obtain the expansion by inserting a resolution of the identity in the primed basis:

$$\varphi_\mu = \sum_\nu u_{\nu\mu}\varphi'_\nu = \left(\sum_\nu |\varphi'_\nu\rangle\langle\varphi'_\nu| \right) |\varphi_\mu\rangle = \sum_\nu \langle\varphi'_\nu|\varphi_\mu\rangle\varphi'_\nu. \tag{5.66}$$

Comparing the second and fourth members of this equation, we identify $u_{\nu\mu}$ as the elements of a matrix U:

$$u_{\nu\mu} = \langle\varphi'_\nu|\varphi_\mu\rangle. \tag{5.67}$$

Note how the use of resolutions of the identity makes these formulas obvious, and that Eqs. (5.65) to (5.67) are only valid because the φ_μ and the φ'_ν are complete orthonormal sets.

Inserting the expansion for φ_μ from Eq. (5.66) into Eq. (5.65), we reach

$$\psi = \sum_\mu c_\mu \sum_\nu u_{\nu\mu}\varphi'_\nu = \sum_\nu \left(\sum_\mu u_{\nu\mu}c_\mu \right) \varphi'_\nu = c'_\nu\varphi'_\nu, \tag{5.68}$$

where the coefficients c'_ν of the expansion in the primed basis form a column vector $\mathbf{c}'$ that is related to the coefficient vector $\mathbf{c}$ in the unprimed basis by the matrix equation

$$\mathbf{c}' = \mathsf{U}\,\mathbf{c}, \tag{5.69}$$

with U the matrix whose elements are given in Eq. (5.67).

If we now consider the reverse transformation, **from** an expansion in the primed basis **to** one in the unprimed basis, starting from

$$\varphi'_\mu = \sum_\nu v_{\nu\mu}\varphi_\nu = \sum_\nu \langle\varphi_\nu|\varphi'_\mu\rangle\varphi_\nu, \tag{5.70}$$

we see that V, the matrix of the transformation inverse to U, has elements

$$v_{\nu\mu} = \langle\varphi_\nu|\varphi'_\mu\rangle = (\mathsf{U}^*)_{\mu\nu} = (\mathsf{U}^\dagger)_{\nu\mu}. \tag{5.71}$$

In other words,

$$\mathsf{V} = \mathsf{U}^\dagger. \tag{5.72}$$

If we now transform the expansion of ψ, given in the unprimed basis by the coefficient vector $\mathbf{c}$, first to the primed basis and then back to the original unprimed basis, the coefficients will transform, first to $\mathbf{c}'$ and then back to $\mathbf{c}$, according to

$$\mathbf{c} = \mathsf{V}\,\mathsf{U}\,\mathbf{c} = \mathsf{U}^\dagger\mathsf{U}\,\mathbf{c}. \tag{5.73}$$

In order for Eq. (5.73) to be consistent it is necessary that $\mathsf{U}^\dagger\mathsf{U}$ be a unit matrix, meaning that U must be **unitary**. We thus have the important following result:

> *The transformation that converts the expansion of a vector* $\mathbf{c}$ *in any orthonormal basis* $\{\varphi_\mu\}$ *to its expansion* $\mathbf{c}'$ *in any other orthonormal basis* $\{\varphi'_\nu\}$ *is described by the matrix equation* $\mathbf{c}' = \mathsf{U}\,\mathbf{c}$*, where the transformation matrix* U *is* **unitary** *and has elements* $u_{\nu\mu} = \langle\varphi'_\nu|\varphi_\mu\rangle$*. A transformation between orthonormal bases is called a* **unitary transformation**.

Equation (5.69) is a direct generalization of the ordinary 2-D vector rotational transformation equation, Eq. (3.26),

$$\mathbf{A}' = \mathsf{S}\mathbf{A}.$$

For further emphasis, we compare the transformation matrix U introduced here (at right, below) with the matrix S (at left) from Eq. (3.28), for rotations in ordinary 2-D space:

$$\mathsf{S} = \begin{pmatrix} \hat{\mathbf{e}}'_1\cdot\hat{\mathbf{e}}_1 & \hat{\mathbf{e}}'_1\cdot\hat{\mathbf{e}}_2 \\ \hat{\mathbf{e}}'_2\cdot\hat{\mathbf{e}}_1 & \hat{\mathbf{e}}'_2\cdot\hat{\mathbf{e}}_2 \end{pmatrix} \qquad \mathsf{U} = \begin{pmatrix} \langle\varphi'_1|\varphi_1\rangle & \langle\varphi'_1|\varphi_2\rangle & \cdots \\ \langle\varphi'_2|\varphi_1\rangle & \langle\varphi'_2|\varphi_2\rangle & \cdots \\ \cdots & \cdots & \cdots \end{pmatrix}.$$

The resemblance becomes even more striking if we recognize that in Dirac notation, the quantities $\hat{\mathbf{e}}'_i\cdot\hat{\mathbf{e}}_j$ assume the form $\langle\hat{\mathbf{e}}'_i|\hat{\mathbf{e}}_j\rangle$.

As for ordinary vectors (except that the quantities involved here are complex), the ith row of U contains the (complex conjugated) components (a.k.a. coefficients) of φ'_i in terms of the unprimed basis; the orthonormality of the primed φ is consistent with the fact that $\mathsf{U}\,\mathsf{U}^\dagger$ is a unit matrix. The columns of U contain the components of the φ_j in terms of the primed basis; that also is analogous to our earlier observations. The matrix S is orthogonal; U is unitary, which is the generalization to a complex space of the orthogonality condition.

Summarizing, we see that unitary transformations are analogous, in vector spaces, to the orthogonal transformations that describe rotations (or reflections) in ordinary space.

Example 5.5.1 A Unitary Transformation

A Hilbert space is spanned by five functions defined on the surface of a unit sphere and expressed in spherical polar coordinates θ, φ:

$$\chi_1 = \sqrt{\frac{15}{4\pi}}\,\sin\theta\cos\theta\cos\varphi, \quad \chi_2 = \sqrt{\frac{15}{4\pi}}\,\sin\theta\cos\theta\sin\varphi,$$

$$\chi_3 = \sqrt{\frac{15}{4\pi}}\,\sin^2\theta\sin\varphi\cos\varphi, \quad \chi_4 = \sqrt{\frac{15}{16\pi}}\,\sin^2\theta(\cos^2\varphi - \sin^2\varphi),$$

$$\chi_5 = \sqrt{\frac{5}{16\pi}}\,(3\cos^2\theta - 1).$$

These are orthonormal when the scalar product is defined as

$$\langle f|g\rangle = \int_0^\pi \sin\theta\,d\theta \int_0^{2\pi} d\varphi\, f^*(\theta,\varphi)\,g(\theta,\varphi).$$

This Hilbert space can alternatively be spanned by the orthonormal set of functions

$$\chi_1' = -\sqrt{\frac{15}{8\pi}}\,\sin\theta\cos\theta\,e^{i\varphi}, \quad \chi_2' = \sqrt{\frac{15}{8\pi}}\,\sin\theta\cos\theta\,e^{-i\varphi},$$

$$\chi_3' = \sqrt{\frac{15}{32\pi}}\,\sin^2\theta\,e^{2i\varphi}, \qquad \chi_4' = \sqrt{\frac{15}{32\pi}}\,\sin^2\theta\,e^{-2i\varphi},$$

$$\chi_5' = \chi_5.$$

The matrix U describing the transformation from the unprimed to the primed basis has elements $u_{\nu\mu} = \langle\chi_\nu'|\chi_\mu\rangle$. Working out a representative matrix element,

$$u_{22} = \langle\chi_2'|\chi_2\rangle = \frac{15}{4\pi\sqrt{2}} \int_0^\pi \sin\theta\,d\theta \int_0^{2\pi} d\varphi\,\sin^2\theta\cos^2\theta\,e^{+i\varphi}\sin\varphi$$

$$= \frac{15}{4\pi\sqrt{2}} \int_0^\pi \sin^3\theta\cos^2\theta\,d\theta \int_0^{2\pi} d\varphi\,e^{+i\varphi}\,\frac{e^{i\varphi}-e^{-i\varphi}}{2i}$$

$$= \frac{15}{4\pi\sqrt{2}}\left(\frac{4}{15}\right)\frac{-2\pi}{2i} = \frac{i}{\sqrt{2}}.$$

We obtained this result by using the formula $\int_0^{2\pi} e^{ni\varphi}\,d\varphi = 2\pi\,\delta_{n0}$ and by looking up a tabulated value for the θ integral. We evaluate explicitly one more matrix element:

$$u_{21} = \langle \chi_2' | \chi_1 \rangle = \frac{15}{4\pi\sqrt{2}} \int_0^{\pi} \sin^3\theta \cos^2\theta \, d\theta \int_0^{2\pi} d\varphi \, e^{+i\varphi} \, \frac{e^{i\varphi} + e^{-i\varphi}}{2}$$

$$= \frac{15}{4\pi\sqrt{2}} \left(\frac{4}{15}\right) \frac{2\pi}{2} = \frac{1}{\sqrt{2}}.$$

Evaluating the remaining elements of U, we reach

$$\mathsf{U} = \begin{pmatrix} -1/\sqrt{2} & -i/\sqrt{2} & 0 & 0 & 0 \\ 1/\sqrt{2} & -i/\sqrt{2} & 0 & 0 & 0 \\ 0 & 0 & i/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & -i/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

As a check, note that the ith column of U should yield the components of χ_i in the primed basis. For the first column, we have

$$\sqrt{\frac{15}{4\pi}} \sin\theta\cos\theta\cos\varphi = -\frac{1}{\sqrt{2}}\left(-\sqrt{\frac{15}{8\pi}} \sin\theta\cos\theta\, e^{i\varphi}\right) + \frac{1}{\sqrt{2}}\left(\sqrt{\frac{15}{8\pi}} \sin\theta\cos\theta\, e^{-i\varphi}\right),$$

which simplifies easily to an identity. Further checks are left as Exercise 5.5.1. ■

Successive Transformations

It is possible to make two or more successive unitary transformations, each of which will convert an input orthonormal basis to an output basis that is also orthonormal. Just as for ordinary vectors, the successive transformations are applied in right-to-left order, and the product of the transformations can be viewed as a resultant unitary transformation.

Exercises

5.5.1 Show that the matrix U of Example 5.5.1 correctly transforms the vector $f(\theta,\varphi) = 3\chi_1 + 2i\chi_2 - \chi_3 + \chi_5$ to the $\{\chi_i'\}$ basis by

(a) (1) Making a column vector $\mathbf{c}$ that represents $f(\theta,\varphi)$ in the $\{\chi_i\}$ basis,

(2) forming $\mathbf{c}' = \mathsf{U}\,\mathbf{c}$, and

(3) comparing the expansion $\sum_i c_i' \chi_i'(\theta,\varphi)$ with $f(\theta,\varphi)$;

(b) Verifying that U is unitary.

5.5.2 (a) Given (in $\mathbb{R}^3$) the basis $\varphi_1 = x$, $\varphi_2 = y$, $\varphi_3 = z$, consider the basis transformation $x \to z,\; y \to y,\; z \to -x$. Find the 3×3 matrix U for this transformation.

(b) This transformation corresponds to a rotation of the coordinate axes. Identify the rotation and reconcile your transformation matrix with an appropriate matrix $S(\alpha, \beta, \gamma)$ of the form given in Eq. (3.37).

(c) Form the column vector $\mathbf{c}$ representing (in the original basis) $f = 2x - 3y + z$, find the result of applying U to $\mathbf{c}$, and show that this is consistent with the basis transformation of part (a).

Note. You do not need to be able to form scalar products to handle this exercise; a knowledge of the linear relationship between the original and transformed functions is sufficient.

5.5.3 Construct the matrix representing the inverse of the transformation in Exercise 5.5.2, and show that this matrix and the transformation matrix of that exercise are matrix inverses of each other.

5.5.4 The unitary transformation U that converts an orthonormal basis $\{\varphi_i\}$ into the basis $\{\varphi_i'\}$ and the unitary transformation V that converts the basis $\{\varphi_i'\}$ into the basis $\{\chi_i\}$ have matrix representations

$$\mathsf{U} = \begin{pmatrix} i\sin\theta & \cos\theta & 0 \\ -\cos\theta & i\sin\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathsf{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & i\sin\theta \\ 0 & \cos\theta & -i\sin\theta \end{pmatrix}.$$

Given the function $f(x) = 3\varphi_1(x) - \varphi_2(x) - 2\varphi_3(x)$,

(a) By applying U, form the vector representing $f(x)$ in the $\{\varphi_i'\}$ basis and then by applying V form the vector representing $f(x)$ in the $\{\chi_i\}$ basis. Use this result to write $f(x)$ as a linear combination of the χ_i.

(b) Form the matrix products UV and VU and then apply each to the vector representing $f(x)$ in the $\{\varphi_i\}$ basis. Verify that the results of these applications differ and that only one of them gives the result corresponding to part (a).

5.5.5 Three functions which are orthogonal with unit weight on the range $-1 \le x \le 1$ are $P_0 = 1$, $P_1 = x$, and $P_2 = \frac{3}{2}x^2 - \frac{1}{2}$. Another set of functions that are orthogonal and span the same space are $F_0 = x^2$, $F_1 = x$, $F_2 = 5x^2 - 3$. Although much of this exercise can be done by inspection, write down and evaluate all the integrals that lead to the results when they are obtained in terms of scalar products.

(a) Normalize each of the P_i and F_i.

(b) Find the unitary matrix U that transforms from the normalized P_i basis to the normalized F_i basis.

(c) Find the unitary matrix V that transforms from the normalized F_i basis to the normalized P_i basis.

(d) Show that U and V are unitary, and that $\mathsf{V} = \mathsf{U}^{-1}$.

(e) Expand $f(x) = 5x^2 - 3x + 1$ in terms of the **normalized** versions of both bases, and verify that the transformation matrix U converts the P-basis expansion of $f(x)$ into its F-basis expansion.

5.6 Transformations of Operators

We have seen how unitary transformations can be used to transform the expansion of a function from one orthonormal basis set to another. We now consider the corresponding transformation for operators. Given an operator A, which when expanded in the φ basis has the form

$$A = \sum_{\mu\nu} |\varphi_\mu\rangle a_{\mu\nu} \langle\varphi_\nu|,$$

we convert it to the φ' basis by the simple expedient of inserting resolutions of the identity (written in terms of the primed basis) on both sides of the above expression. This is an excellent example of the benefits of using Dirac notation. Remembering that this does not change A (but of course does change its appearance), we get

$$A = \sum_{\mu\nu\sigma\tau} |\varphi'_\sigma\rangle\langle\varphi'_\sigma|\varphi_\mu\rangle a_{\mu\nu}\langle\varphi_\nu|\varphi'_\tau\rangle\langle\varphi'_\tau|,$$

which we simplify by identifying $\langle\varphi'_\sigma|\varphi_\mu\rangle = u_{\sigma\mu}$, as defined in Eq. (5.67), and $\langle\varphi_\nu|\varphi'_\tau\rangle = u^*_{\tau\nu}$. Thus,

$$A = \sum_{\mu\nu\sigma\tau} |\varphi'_\sigma\rangle u_{\sigma\mu} a_{\mu\nu} u^*_{\tau\nu}\langle\varphi'_\tau| = \sum_{\sigma\tau} |\varphi'_\sigma\rangle a'_{\sigma\tau}\langle\varphi'_\tau|, \tag{5.74}$$

where $a'_{\sigma\tau}$ is the $\sigma\tau$ matrix element of A in the primed basis, related to the unprimed values by

$$a'_{\sigma\tau} = \sum_{\mu\nu} u_{\sigma\mu} a_{\mu\nu} u^*_{\tau\nu}. \tag{5.75}$$

If we now note that $u^*_{\tau\nu} = (\mathsf{U}^\dagger)_{\nu\tau}$, we can write Eq. (5.75) as the matrix equation

$$\mathsf{A}' = \mathsf{U}\,\mathsf{A}\,\mathsf{U}^\dagger = \mathsf{U}\,\mathsf{A}\,\mathsf{U}^{-1}, \tag{5.76}$$

where in the final member of the equation we used the fact that U is unitary.

Another way of getting at Eq. (5.76) is to consider the operator equation $A\psi = \chi$, where initially A, ψ, and χ are all regarded as expanded in the orthonormal set φ, with A having matrix elements $a_{\mu\nu}$, and with ψ and χ having the forms $\psi = \sum_\nu c_\mu \varphi_\mu$ and $\chi = \sum_\nu b_\nu \varphi_\nu$. This state of affairs corresponds to the matrix equation

$$\mathsf{A}\,\mathbf{c} = \mathbf{b}.$$

Now we simply insert $\mathsf{U}^{-1}\mathsf{U}$ between A and $\mathbf{c}$, and multiply both sides of the equation on the left by U. The result is

$$\left(\mathsf{U}\,\mathsf{A}\,\mathsf{U}^{-1}\right)\left(\mathsf{U}\,\mathbf{c}\right) = \mathsf{U}\,\mathbf{b} \quad\longrightarrow\quad \mathsf{A}'\mathbf{c}' = \mathbf{b}', \tag{5.77}$$

showing that the operator and the functions are properly related when the functions have been transformed by applying U and the operator has been transformed as required by Eq. (5.76). Since this relationship is valid for any choice of $\mathbf{c}$ and U, it confirms the transformation equation for A.

Nonunitary Transformations

It is possible to consider transformations similar to that illustrated by Eq. (5.77), but using a transformation matrix G that must be nonsingular, but is not required to be unitary. Such more general transformations occasionally appear in physics applications, are called **similarity transformations**, and lead to an equation deceptively similar to Eq. (5.77):

$$\left(\mathsf{G}\,\mathsf{A}\,\mathsf{G}^{-1}\right)\left(\mathsf{G}\mathbf{c}\right) = \mathsf{G}\mathbf{b}. \tag{5.78}$$

There is one important difference: Although a general similarity transformation preserves the original operator equation, corresponding items do not describe the same quantity in a different basis. Instead, they describe quantities that have been systematically (but consistently) altered by the transformation.

Sometimes we encounter a need for transformations that are not even similarity transformations. For example, we may have an operator whose matrix elements are given in a nonorthogonal basis, and we consider the transformation to an orthonormal basis generated by use of the Gram-Schmidt procedure.

Example 5.6.1 Gram-Schmidt Transformation

The Gram-Schmidt process describes the transformation from an initial function set χ_i to an orthonormal set φ_μ according to equations that can be brought to the form

$$\varphi_\mu = \sum_{i=1}^{\mu} t_{i\mu}\chi_i, \qquad \mu = 1, 2, \ldots .$$

Because the Gram-Schmidt process only generates coefficients $t_{i\mu}$ with $i \le \mu$, the transformation matrix T can be described as **upper triangular**, i.e., a square matrix with nonzero elements $t_{i\mu}$ only on and above its principal diagonal. Defining S as a matrix with elements $s_{ij} = \langle\chi_i|\chi_j\rangle$ (often called an **overlap** matrix), the orthonormality of the φ_μ is evidenced by the equation

$$\langle\varphi_\mu|\varphi_\nu\rangle = \sum_{ij}\langle t_{i\mu}\chi_i|t_{j\nu}\chi_j\rangle = \sum_{ij} t^*_{i\mu}\langle\chi_i|\chi_j\rangle t_{j\nu} = (\mathsf{T}^\dagger\mathsf{S}\mathsf{T})_{\mu\nu} = \delta_{\mu\nu}. \tag{5.79}$$

Note that because T is upper triangular, $\mathsf{T}^\dagger$ must be **lower triangular**. In writing Eq. (5.79) we did not have to restrict the i and j summations, as the coefficients outside the contributing ranges of i and j are present, but set to zero.

From Eq. (5.79) we can obtain a representation of S:

$$\mathsf{S} = (\mathsf{T}^\dagger)^{-1}\mathsf{T}^{-1} = (\mathsf{T}\mathsf{T}^\dagger)^{-1}. \tag{5.80}$$

Moreover, if we replace S from Eq. (5.79) by the matrix of a general operator A (in the χ_i basis), we find that in the orthonormal φ basis its representation A' is

$$\mathsf{A}' = \mathsf{T}^\dagger\mathsf{A}\mathsf{T}. \tag{5.81}$$

In general, $\mathsf{T}^\dagger$ will not be equal to T^{-1}, so this equation does not define a similarity transformation. ■

Exercises

5.6.1 (a) Using the two spin functions $\varphi_1 = \alpha$ and $\varphi_2 = \beta$ as an orthonormal basis (so $\langle\alpha|\alpha\rangle = \langle\beta|\beta\rangle = 1$, $\langle\alpha|\beta\rangle = 0$), and the relations

$$S_x\alpha = \frac{1}{2}\beta,\quad S_x\beta = \frac{1}{2}\alpha,\quad S_y\alpha = \frac{1}{2}i\beta,\quad S_y\beta = -\frac{1}{2}i\alpha,\quad S_z\alpha = \frac{1}{2}\alpha,\quad S_z\beta = -\frac{1}{2}\beta,$$

construct the 2×2 matrices of S_x, S_y, and S_z.

(b) Taking now the basis $\varphi'_1 = C(\alpha + \beta)$, $\varphi'_2 = C(\alpha - \beta)$:

(i) Verify that φ'_1 and φ'_2 are orthogonal,

(ii) Assign C a value that makes φ'_1 and φ'_2 normalized,

(iii) Find the unitary matrix for the transformation $\{\varphi_i\} \to \{\varphi'_i\}$.

(c) Find the matrices of S_x, S_y, and S_z in the $\{\varphi'_i\}$ basis.

5.6.2 For the basis $\varphi_1 = Cxe^{-r^2}$, $\varphi_2 = Cye^{-r^2}$, $\varphi_3 = Cze^{-r^2}$, where $r^2 = x^2 + y^2 + z^2$, with the scalar product defined as an unweighted integral over $\mathbb{R}^3$ and with C chosen to make the φ_i normalized:

(a) Find the 3×3 matrix of $L_x = -i\left(y\dfrac{\partial}{\partial z} - z\dfrac{\partial}{\partial y}\right)$;

(b) Using the transformation matrix $\mathsf{U} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -i/\sqrt{2} \\ 0 & 1/\sqrt{2} & i/\sqrt{2} \end{pmatrix}$, find the transformed matrix of L_x;

(c) Find the new basis functions φ'_i defined by the transformation U, and write explicitly (in terms of x, y, and z) the functional forms of $L_x\varphi'_i$, $i = 1,\ 2,\ 3$.

Hint. Use $\int e^{-r^2} d^3r = \pi^{3/2}$, $\int x^2 e^{-r^2} d^3r = \frac{1}{2}\pi^{3/2}$; the integrals are over $\mathbb{R}^3$.

5.6.3 The Gram-Schmidt process for converting an arbitrary basis χ_ν into an orthonormal set φ_ν is described in Section 5.2 in a way that introduces coefficients of the form $-\langle\varphi_\mu|\chi_\nu\rangle$. For bases consisting of three functions, convert the formulation so that φ_ν is expressed entirely in terms of the χ_μ, thereby obtaining an expression for the upper-triangular matrix T appearing in Eq. (5.81).

5.7 INVARIANTS

Just as coordinate rotations leave invariant the essential properties of physical vectors, we can expect unitary transformations to preserve essential features of our vector spaces. These invariances are most directly observed in the basis-set expansions of operators and functions.

Consider first a matrix equation of the form $\mathbf{b} = \mathsf{A}\mathbf{c}$, where all quantities have been evaluated using a particular orthonormal basis φ_i. Now suppose that we wish to use a basis χ_i which can be reached from the original basis by applying a unitary transformation such that

$$\mathbf{c}' = \mathsf{U}\,\mathbf{c} \quad \text{and} \quad \mathbf{b}' = \mathsf{U}\,\mathbf{b}.$$

In the new basis, the matrix A becomes $\mathsf{A}' = \mathsf{UAU}^{-1}$, and the invariance we seek corresponds to $\mathbf{b}' = \mathsf{A}'\mathbf{c}'$. In other words, all the quantities must change coherently so that their relationship is unaltered. It is easy to verify that this is the case. Substituting for the primed quantities,

$$\mathsf{U}\,\mathbf{b} = (\mathsf{UAU}^{-1})(\mathsf{U}\,\mathbf{c}) \quad \longrightarrow \quad \mathsf{U}\,\mathbf{b} = \mathsf{U}\,\mathsf{A}\,\mathbf{c},$$

from which we can recover $\mathbf{b} = \mathsf{A}\mathbf{c}$ by multiplying from the left by U^{-1}.

Scalar quantities should remain invariant under unitary transformation; the prime example here is the scalar product. If f and g are represented in some orthonormal basis, respectively, by $\mathbf{a}$ and $\mathbf{b}$, their scalar product is given by $\mathbf{a}^\dagger\mathbf{b}$. Under a unitary transformation whose matrix representation is U, $\mathbf{a}$ becomes $\mathbf{a}' = \mathsf{U}\,\mathbf{a}$ and $\mathbf{b}$ becomes $\mathbf{b}' = \mathsf{U}\,\mathbf{b}$, and

$$\langle f|g\rangle = (\mathbf{a}')^\dagger\mathbf{b}' = (\mathsf{U}\,\mathbf{a})^\dagger(\mathsf{U}\,\mathbf{b}) = (\mathbf{a}^\dagger\mathsf{U}^\dagger)(\mathsf{U}\,\mathbf{b}) = \mathbf{a}^\dagger\mathbf{b}. \tag{5.82}$$

The fact that $\mathsf{U}^\dagger = \mathsf{U}^{-1}$ enables us to confirm the invariance.

Another scalar that should remain invariant under basis transformation is the expectation value of an operator.

Example 5.7.1 Expectation Value in Transformed Basis

Suppose that $\psi = \sum_i c_i\varphi_i$, and that we wish to compute the expectation value of A for this ψ, where A, the matrix corresponding to A, has elements $a_{\nu\mu} = \langle\varphi_\nu|A|\varphi_\mu\rangle$. We have

$$\langle A\rangle = \langle\psi|A|\psi\rangle \quad \longrightarrow \quad \mathbf{c}^\dagger\mathsf{A}\mathbf{c}.$$

If we now choose to use a basis obtained from the φ_i by a unitary transformation U, the expression for $\langle A\rangle$ becomes

$$(\mathsf{U}\,\mathbf{c})^\dagger(\mathsf{UAU}^{-1})(\mathsf{U}\,\mathbf{c}) = \mathbf{c}^\dagger\mathsf{U}^\dagger\mathsf{UAU}^{-1}\mathsf{U}\,\mathbf{c},$$

which, because U is unitary and therefore $\mathsf{U}^\dagger = \mathsf{U}^{-1}$, reduces, as it must, to the previously obtained value of $\langle A\rangle$. ■

Vector spaces have additional useful matrix invariants. The **trace** of a matrix is invariant under unitary transformation. If $\mathsf{A}' = \mathsf{U}\,\mathsf{A}\,\mathsf{U}^{-1}$, then

$$\begin{aligned}\operatorname{trace}(\mathsf{A}') &= \sum_\nu(\mathsf{U}\,\mathsf{A}\,\mathsf{U}^{-1})_{\nu\nu} = \sum_{\nu\mu\tau} u_{\nu\mu}a_{\mu\tau}(\mathsf{U}^{-1})_{\tau\nu} = \sum_{\mu\tau}\left(\sum_\nu(\mathsf{U}^{-1})_{\tau\nu}u_{\nu\mu}\right)a_{\mu\tau}\\ &= \sum_{\mu\tau}\delta_{\mu\tau}a_{\mu\tau} = \sum_\mu a_{\mu\mu} = \operatorname{trace}(\mathsf{A}).\end{aligned} \tag{5.83}$$

Here we simply used the property $\mathsf{U}^{-1}\mathsf{U} = \mathbf{1}$.

Another matrix invariant is the determinant. From the determinant product theorem, $\det(\mathsf{U}\,\mathsf{A}\,\mathsf{U}^{-1}) = \det(\mathsf{U}^{-1}\mathsf{U}\,\mathsf{A}) = \det(\mathsf{A})$. Further invariants will be identified when we study matrix eigenvalue problems in Chapter 6.

Exercises

5.7.1 Using the formal properties of unitary transformations, show that the commutator $[x, p] = i$ is invariant under unitary transformation of the matrices representing x and p.

5.7.2 The Pauli matrices

$$\boldsymbol{\sigma}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \boldsymbol{\sigma}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \boldsymbol{\sigma}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

have commutator $[\boldsymbol{\sigma}_1, \boldsymbol{\sigma}_2] = 2i\boldsymbol{\sigma}_3$. Show that this relationship continues to be valid if these matrices are transformed by

$$\mathsf{U} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

5.7.3 (a) The operator L_x is defined as

$$L_x = -i\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right).$$

Verify that the basis $\varphi_1 = Cxe^{-r^2}$, $\varphi_2 = Cye^{-r^2}$, $\varphi_3 = Cze^{-r^2}$, where $r^2 = x^2 + y^2 + z^2$, forms a closed set under the operation of L_x, meaning that when L_x is applied to any member of this basis the result is a function within the basis space, and construct the 3×3 matrix of L_x in this basis from the result of the application of L_x to each basis function.

(b) Verify that $L_x\left[(x + iy)e^{-r^2}\right] = -ze^{-r^2}$, and note that this result, using the $\{\varphi_i\}$ basis, can be written $L_x(\varphi_1 + i\varphi_2) = -\varphi_3$.

(c) Express the equation of part (b) in matrix form, and write the matrix equation that results when each of the quantities is transformed using the transformation matrix

$$\mathsf{U} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -i/\sqrt{2} \\ 0 & 1/\sqrt{2} & i/\sqrt{2} \end{pmatrix}.$$

(d) Regarding the transformation U as producing a new basis $\{\varphi_i'\}$, find the explicit form (in x, y, z) of the φ_i'.

(e) Using the operator form of L_x and the explicit forms of the φ_i', verify the validity of the transformed equation found in part (c).

Hint. The results of Exercise 5.6.2 may be useful.

5.8 Summary—Vector Space Notation

It may be useful to summarize some of the relationships found in this chapter, highlighting the essentially complete mathematical parallelism between the properties of vectors and those of basis expansions in vector spaces. We do so here, using Dirac notation wherever appropriate.

1. **Scalar product:**

$$\langle \varphi|\psi\rangle = \int_a^b \varphi^*(t)\psi(t)w(t)\,dt \Longleftrightarrow \langle \mathbf{u}|\mathbf{v}\rangle = \mathbf{u}^\dagger \mathbf{v} = \mathbf{u}^* \cdot \mathbf{v}. \tag{5.84}$$

The result of the scalar product operation is a scalar (i.e., a real or complex number). Here $\mathbf{u}^\dagger\mathbf{v}$ represents the product of a row and a column vector; it is equivalent to the dot-product notation also shown.

2. **Expectation value:**

$$\langle \varphi|A|\varphi\rangle = \int_a^b \varphi^*(t)A\varphi(t)w(t)\,dt \Longleftrightarrow \langle \mathbf{u}|\mathsf{A}|\mathbf{u}\rangle = \mathbf{u}^\dagger \mathsf{A}\mathbf{u}. \tag{5.85}$$

3. **Adjoint:**

$$\langle \varphi|A|\psi\rangle = \langle A^\dagger\varphi|\psi\rangle \Longleftrightarrow \langle \mathbf{u}|\mathsf{A}|\mathbf{v}\rangle = \langle \mathsf{A}^\dagger\mathbf{u}|\mathbf{v}\rangle = [\mathsf{A}^\dagger\mathbf{u}]^\dagger\mathbf{v} = \mathbf{u}^\dagger\mathsf{A}\mathbf{v}. \tag{5.86}$$

Note that the simplification of $[\mathsf{A}^\dagger\mathbf{u}]^\dagger\mathbf{v}$ shows that the matrix $\mathsf{A}^\dagger$ has the property expected of an operator adjoint.

4. **Unitary transformation:**

$$\psi = A\varphi \longrightarrow U\psi = (UAU^{-1})(U\varphi) \Longleftrightarrow \mathbf{w} = \mathsf{A}\mathbf{v} \longrightarrow \mathsf{U}\mathbf{w} = (\mathsf{U}\mathsf{A}\mathsf{U}^{-1})(\mathsf{U}\mathbf{v}). \tag{5.87}$$

5. **Resolution of identity:**

$$\mathbf{1} = \sum_i |\varphi_i\rangle\langle\varphi_i| \Longleftrightarrow \mathbf{1} = \sum_i |\hat{\mathbf{e}}_i\rangle\langle\hat{\mathbf{e}}_i|, \tag{5.88}$$

where the φ_i are orthonormal and the $\hat{\mathbf{e}}_i$ are orthogonal unit vectors. Applying Eq. (5.88) to a function (or vector):

$$\psi = \sum_i |\varphi_i\rangle\langle\varphi_i|\psi\rangle = \sum_i a_i\varphi_i \Longleftrightarrow \mathbf{w} = \sum_i |\hat{\mathbf{e}}_i\rangle\langle\hat{\mathbf{e}}_i|\mathbf{w}\rangle = \sum_i w_i\hat{\mathbf{e}}_i, \tag{5.89}$$

where $a_i = \langle\varphi_i|\psi\rangle$ and $w_i = \langle\hat{\mathbf{e}}_i|\mathbf{w}\rangle = \hat{\mathbf{e}}_i \cdot \mathbf{w}$.

Additional Readings

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